

Decomposing epistemic verification: an operational account of expert judgment and its limits*

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Abstract

Much of what professional settings call “judgment,” “critical thinking,” “senior review,” or “the expert eye” is epistemic verification: the work of checking whether a claim can actually bear the weight placed on it. That practice decomposes. It separates into ten named operations (crux identification, controlling-claim isolation, reframing recognition, charter-drift detection, anchor-proxy requirement, directed search, failure-mode tagging, promissory evidence detection, quarantine-move detection, fail-closed defaulting), twelve recurring pathologies the operations are designed to catch, and seven process principles that keep the decomposition auditable under optimization pressure (separation of generation from verification, statelessness without social cost, typed operations bound to deterministic checks, cheap repetition, pre-registration of the killer question, holdout test surfaces authored outside the candidate’s claim region, and asymptotic scoring standards). An empirical mining of 2,395 scored iterations across 121 projects shows that the failure-mode taxonomy splits into two causal categories: structural blockers (circularity, harness defects, unfalsifiable claims) that are incompatible with high scores, and ceiling-breakers (missing counterfactuals, overclaimed scope, catastrophic assumptions) that are unexpectedly more common at high scores than at low ones. The two categories call for opposite prompt-design responses. A residual remains: three operations (selection of the crux itself, recognition of when a reframing is required, the social dynamics of live pressure-testing) that did not decompose in the system studied. The decomposition is proposed abductively from one recursive adversarial verification system over roughly one month. The claim is bounded and falsifiable: it would fail if the operations did not remain separable under independent replication, or if the named pathologies did not recur outside the present corpus. Generalization beyond this system awaits independent replication.

Front matter: scope and orientation

Hand an experienced reviewer a confident, well-formatted analysis, a memo that values an acquisition on a single buried cost line, say, or a model whose headline number turns on one unstated parameter, and watch what happens. They do not read it top to bottom. They find the load-bearing claim, the assumption the whole result rests on, and press there. Often they cannot say how they knew where to press. The whole performance gets filed under one word: judgment.

Much of what we call judgment, critical thinking, senior review, or the expert eye is *epistemic verification*: the work of checking whether a claim can bear the weight placed on it. That work is not one faculty. It decomposes into named operations, and several of them can be made procedural

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and auditable, carried out by a process whose incentives are separated from whoever produced the claim.

The decomposition does not close. Three operations resist it, and their residual matters as much as the operations that decompose. Where verification can be made procedural, it should be, where it cannot, that boundary is where judgment lives, smaller and better located than the one word lets on.

Separating a generator from its evaluator implies an organizational architecture, which a companion paper works out (*The Cognitive Firm*, Alami 2026c). Here the subject is narrower: the verification operations that make such separation worth its cost.

The evidence is a single case, and that is the largest limitation of this paper. Everything below comes from one system I built and operated alone over about a month, and every claim that follows is bounded by that fact. I drew the ten operations and seven principles abductively from one recursive adversarial verification system, which I operated alone over about a month. A single case can establish that a mechanism exists and specify its structure, leaving its prevalence to replication (Yin 2018, Glaser and Strauss 1967), Ebbinghaus charted the forgetting curve on one subject, and patient H.M. grounded the modern theory of distinct memory systems on the same logic. The operations are offered for others to test on other systems. Throughput, cost per validated finding, and comparative efficiency wait for later instrumentation, the claim here is structural, about what the operations are. (One data-integrity note: some early runs predate a fully sealed tool-use corridor, a sweep of 175 debate logs found no scraping patterns, raw tool output was not retained so the absence cannot be proven retroactively, and runs after 2026-04-15 are clear.)

Taylor anchors the method because scientific management made one move worth borrowing: it refused to treat skilled practice as ineffable because no one had yet decomposed it. I borrow that methodological move, decompose the practice, name the operations, not Taylor's politics, labor regime, or distributional conclusions.

One word, *agency*, does two different jobs here, and collapsing them makes the decomposition look as if it threatens something it does not. Jensen and Meckling (1976) use *agency* for the agency problem under misaligned incentives (the structural sense the companion paper *The Cognitive Firm* uses). Bandura (1989) and Ryan and Deci (2000) use *agency* for the psychological capacity to originate action under self-endorsed motivation. Chapters 1 and 2 make a claim about the first sense: the operations can be performed by a process whose incentives are structurally separated from the generator's. The residual chapter makes a claim about the second sense: the three operations that did not decompose are the ones whose value depends on being performed by a causal originator under self-endorsed motivation. Keeping the two senses distinct is a prerequisite for arguing about what stays human.

Peirce clarifies why abductive framing remains hard to mechanize. Campbell, Popper, and Hull clarify why a generator needs a selection environment. Drucker is a reminder that systematization leaves a managerial and human residual. The proof burden is operational, not borrowed from these references: whether the proposed operations identify real verification moves and recurring failures.

This paper contributes an operational decomposition of epistemic verification into ten named, auditable operations, paired with the recurring pathologies they catch and the process principles that keep them usable under optimization pressure. It also names a residual, the operations that resist decomposition, and treats that boundary as a finding, not a failure of analysis.

Introduction: judgment as verification

Many institutions depend on a practice they cannot describe very well. A senior reviewer reads a memo and sees that the argument will not hold. A referee rejects a paper because the evidence answers the wrong question. A diligence lead hears a persuasive story and asks for the one observation that would falsify it. These are verification moves, usually learned by apprenticeship and usually described after the fact as judgment.

The practice decomposes into named operations, a set of process principles that keep those operations auditable under optimization pressure, and a residual that has not decomposed in the system studied here. The whole comes from one working system, and replication elsewhere must come before it generalizes.

One objection is historical and ethical. Scientific management often reduced autonomy and shifted the gains of decomposition away from the workers whose practice was decomposed. That history matters, because a decomposition of epistemic work could be used the same way.

That objection is serious, and the residual chapter answers it directly by naming what the decomposition does not capture. A decomposition makes it easier to say which parts of the work should be proceduralized, which should remain human responsibilities, and where the boundary is still uncertain.

A second objection is methodological: epistemic work may not decompose without losing the property that made it valuable. That objection controls everything else, and the decomposition answers it by asking whether the named operations correspond to moves readers recognize in actual review, research, diligence, and governance work.

A decomposition fails if it does not name moves a practitioner recognizes. If it does, the live questions are where those moves can be made procedural, where they remain semantic and fallible, and where human responsibility still enters.

Neither “LLMs produce answers and humans inspect them” nor ordinary domain-expert research describes the system studied here very well. It is a four-role operating loop. First, a generator proposes a candidate object: a claim, law, theorem packet, bridge, discriminator, or failure explanation. Second, deterministic and algebraic machinery close whatever local formal vulnerability can be closed cheaply: exact arithmetic, symbolic identities, held-out fits, typed gates, and other bounded checks. Third, a compression or classification pass asks what family the surviving object actually belongs to, so that a local success is not mistaken for a stronger category than it has earned. Fourth, a human reviewer acts as capital allocator and promotion gate, advancing an object only if it materially changes the live belief graph or decision graph and demoting it when the next hostile check breaks it. The decomposition in the chapters that follow is a claim about the verification and promotion parts of that loop. The loop is the operational setting in which those parts became visible.

Related work

One tradition treats expert judgment as tacit and resistant to articulation. Polanyi (1966) holds that we know more than we can tell. Dreyfus and Dreyfus (1986) argue that expert performance runs on situational intuition, not on rules a novice could be handed. Klein (1998) documents the recognition-primed decisions practitioners make under time pressure, and the expert-performance

literature after Ericsson treats skill as accumulated pattern more than statable procedure. This paper bets against that view on part of the territory: much of verification is articulable as named operations bound to checks, while Chapter 3 concedes a residual that genuinely stays tacit.

A second tradition systematizes the structure of arguments more than the work of checking them. Toulmin (1958) parses an argument into claim, warrant, and backing. Informal logic (Walton 1989) catalogues the fallacies a critic looks for. Both describe what an argument is made of and what can go wrong with it in the abstract. Neither models what a verifier does to a live argument under sustained optimization pressure, where the thing being checked adapts to the check. That dynamic is the subject here.

Closest in spirit is recent work on machine oversight: AI safety via debate (Irving et al. 2018), reward modeling (Leike et al. 2018), measured scalable oversight (Bowman et al. 2022), weak-to-strong generalization (Burns et al. 2023), and the specification-gaming catalogue (Krakovna et al. 2020). That work builds and stress-tests the overseer and asks whether it holds as the supervised system grows stronger. It treats the verifier largely as one mechanism. This paper opens that mechanism and asks which of its operations decompose into auditable checks and which do not, the question the oversight literature leaves implicit.

Between the tacit-knowledge tradition, which holds that the work cannot be decomposed, and the oversight tradition, which mechanizes the verifier without decomposing it, this paper takes the middle position: name the operations that decompose, and mark the ones that do not.

Chapter 1: the decomposition

What gets decomposed

At issue is *epistemic verification*: taking a claim, an argument, or a proposed decision and checking whether it actually supports the weight being placed on it. It sits apart from three adjacent activities it is routinely confused with.

Verification stands downstream of *generation*, the production of claims, arguments, or proposals. Most published metrics about AI in knowledge work measure generation throughput: how fast a system produces documents, how fluent the documents are, how much human writing time it displaces. These metrics are real, but none of them measures verification.

Verification sits upstream of *decision-making*, committing to a course of action under uncertainty. Decision-making is an executive function performed on verified or unverified inputs. The verification work separates from the commitment work, and a different party or process can perform it.

Verification is also distinct from *analysis*, producing structured representations of a situation. Analysis itself remains susceptible to the pathologies that verification is designed to catch. An unverified analysis is something to verify, never a substitute for having verified it.

Epistemic verification asks two linked questions of a claim: *what would have to be true for it to hold, and is that the case?* It is the operation a senior partner performs when they read a junior’s memo and ask “what is controlling here?” It is the operation a case-method instructor performs when they force the protagonist’s story into contact with its hidden assumptions. It is the operation a good referee performs on a submitted manuscript. It is the operation a diligence lead performs on a pitch deck. It is the thing that, when it has been done well, produces the feeling that an argument

has survived pressure, and, when it has been done poorly or not at all, produces the feeling that an argument is smooth enough to win the room without being sound enough to win reality.

The ten operations

Observed in an adversarial verification system running across multiple domains over roughly one month, epistemic verification decomposes into the operations below. Several adapt ideas with a longer lineage: the crux echoes Collingwood’s logic of question and answer and Polya’s stress on finding the unknown, reframing draws on Schön and Kuhn, the ad hoc rescue and promissory evidence on Popper’s immunizing stratagems and Lakatos’s monster-barring, the operational proxy on Bridgman’s operationalization, and separating generation from verification on Reichenbach’s distinction between the contexts of discovery and justification. What is new is the decomposition into auditable operations. The individual notions are borrowed. They are listed as a flat set. In practice they compose and recur, and that composition is a property of the work.

1. *Crux identification.* Any verification pass begins by identifying the single question whose answer determines the answer to every other question in the set. Most arguments contain a dozen claims, a decomposed pass finds the one claim whose truth or falsity collapses the rest and attacks that one first, before working through the others. Identifying it is learnable, there is a move-set for recognizing which claim is the controlling claim, and even intelligent readers do not do it automatically. They routinely spend effort on peripheral claims, because the vocabulary of “critical thinking” does not distinguish attacking any claim from attacking the right one. One terminological distinction matters here. *Crux identification* (this operation) takes a given argument and finds the controlling claim within it. *Crux selection*, treated in the residual chapter, is the prior choice of which argument to verify in the first place. Identification decomposes, selection is residual. They share a name because they share a structure, but they differ on the axis that defines the residual: identification operates on a structured input, while selection operates before the input exists.

2. *Controlling claim isolation.* Once the crux has been identified, isolate the claim that carries the argument’s weight, the claim that, if removed, causes the downstream structure to fall. This is a distinct operation from crux identification. A crux is the question the argument is supposed to answer, the controlling claim is the part of the argument doing the work of answering it. They are not always the same. An argument can have a clean crux and yet depend on a premise that looks peripheral.

3. *Reframing recognition.* Some verification problems cannot be solved by attacking the argument as it stands. They require a change in the *shape* of the argument: a re-wiring of the dependency graph rather than a refutation of any particular node. Recognizing when the required move is a reframing is a distinct operation, difficult, and often absent from unstructured critique. A verifier who misses it spends unbounded effort attacking individual claims in an argument whose structure is the actual problem.

4. *Scope drift detection.* Arguments, especially ones produced under time pressure, tend to migrate from their stated scope to an adjacent, more comfortable scope. The argument answers a question, but not always the question it was asked to answer. Scope drift detection compares the question the argument is actually answering against the question it was commissioned to answer, a routine move in well-run diligence and a rare one outside of it.

5. *Operational proxy requirement.* When an argument relies on a variable that has no observable proxy (“political will,” “organizational alignment,” “underlying demand”) requiring an operational

proxy means insisting that the abstract variable bind to something a third party could read off a dashboard on a given day. A proxy need not be perfect, it only has to exist. An argument that cannot produce one on request rests on a controlling variable that is a ghost metric in waiting.

6. *Directed search.* When a thesis fails under pressure, directed search explores whether a better thesis exists within the current framing, short of the harder move of reframing. It is the cheap option and the correct first move when a thesis has failed for a local, recoverable reason. It is the wrong move when the failure is structural, where the correct operation is the reframing of Operation 3. A verifier who does not distinguish the two cases either reframes too quickly, discarding recoverable theses, or persists too long, running directed search inside a framing that is the actual problem.

7. *Failure-mode tagging.* Assigning a canonical name to a recurring structural failure so that it can be recognized in new contexts. The naming is what turns a one-off incident into reusable precedent. Without it, each verification pass rediscovers the same failure modes from scratch and the practice does not accumulate. The ten field manual patterns catalogued below are what sustained failure-mode tagging across domains produced.

8. *Promissory evidence detection.* Catching the move where the evidence for a claim is pushed into the future, and then the plan to seek that evidence is treated as if the evidence were already in hand, the review-process cousin of the promissory evidence (Pattern 1 in the catalogue below). It is a specific operation because it requires the verifier to hold, simultaneously, the current state of evidence and the rhetorical treatment of that evidence, and to notice when the two have drifted.

9. *Ad hoc rescue detection.* Catching the move where a risk or a failure is isolated into a labeled section of the argument and then the rest of the argument proceeds as if the labeling itself were mitigation. The same move sits underneath Pattern 3 of the field manual (acknowledged but unmitigated risk). Detection differs from pattern recognition in what it tracks: not just the acknowledgment of the risk but the downstream treatment of the acknowledgment.

10. *Fail-closed defaulting.* Returning *failure* when an operation hits an unexpected state, instead of guessing or continuing. It is the only operation on this list with a direct parallel in engineering practice: the same fail-closed default a well-designed safety system uses when its sensors disagree. In epistemic verification, it refuses to convert ambiguity into false certainty by default, “I do not know” is the correct answer until the evidence is actually present.

Epistemic modesty derives from the evidence boundary. A thesis that states “tail class unresolved” because its farther-tail gate returned a residual five times above threshold is computing its uncertainty from the apparatus. A thesis that states “this might be uncertain” with no corresponding gate failure is performing compliance. The apparatus enforces the difference by construction: pre-registration (Principle V) fixes the criterion before the argument exists, and the gate output is the derivation that makes the hedge controlling rather than decorative. Fail-closed defaulting is the operation most incumbent critical-thinking vocabularies have no name for, because the incumbent practice leans on its opposite: producing a confident-sounding answer when the evidence is not yet sufficient.

Ten operations make a decomposition, and wider domains may surface more. The list is not final, but it names observable distinctions: each operation describes a move that can be taught, looked for, and audited against characteristic failures.

The twelve pathologies

Each operation above is defined in part by the failures it is designed to catch. The operations exist in response to recurring failure modes that arguments exhibit under pressure, so a decomposition that named the operations without naming those failures would be incomplete. The twelve patterns below are the pathologies the operations detect, the empirical output of a structured adversarial verification process applied to four or five projects across three distinct strategic domains. Their provenance varies: some appear in a single domain (“Tentative”), some across two (“Probable”), and one across three or more (“Confirmed”). A versioned field manual holds the full catalogue, abridged below.

Each pathology is an artifact-level example of deeper failure dynamics documented in the author’s prior work on epistemic supervision (Alami 2026b). When a generator can optimize against the evaluator’s decision surface, it learns shortcuts through the evaluation. The twelve patterns below are recurring shapes those shortcuts take, a field manual for inspection rather than a complete theory of reasoning failure.

Lakatos’s distinction between progressive and degenerating research programmes (Lakatos 1978) explains why several patterns below are dangerous: they protect a thesis by fitting the known case where a progressive move would risk a novel observable. Ashby’s Law of Requisite Variety (Ashby 1956) supplies the engineering analogue: a verifier whose categories are poorer than the generator’s moves will eventually be out-dimensioned. The process principles below state the response to that constraint.

Several of these are long-established failures, and the names here are the literature’s, not coinages. The persuasive definition is Stevenson’s, the conclusion made true by how the question is drawn. The false dilemma is the classic informal fallacy. The category error is Ryle’s. The unmeasurable construct restates Bridgman’s operationalist demand that a construct bind to an observable. The unfalsifiable forecast is Popper’s. Non-diagnostic evidence is the Bayesian case of a likelihood ratio near one. The catalogue’s work is to pair each failure with the operation that catches it.

Promissory evidence. [Confirmed: observed in startup strategy, scientific law recovery, and military intelligence domains.] An argument claims its central proof exists, but the proof is something that hasn’t happened yet. A reader nods because the logic *would* work, if the future event arrives in the predicted shape. Ask the killer question: *what observation available to us this week would change your mind about this?*

Non-diagnostic evidence. [Probable: observed in startup strategy and scientific law recovery domains.] An argument points to a piece of evidence and says “this is what makes my theory right,” but the same evidence fits the rival theory just as well. The metric does not discriminate, it only sounds discriminating. Here the killer question runs: *what would the world have to look like for this same number to appear even if your hypothesis were wrong?*

Acknowledged but unmitigated risk. [Tentative: observed in a single domain (startup strategy).] A fatal risk is named explicitly, almost gracefully, and then the argument continues as if naming it had retired it. An audience interprets the acknowledgment as rigor. Ask, then: *if the risk you just named landed in the worst plausible form, which numbers in the rest of this deck stop being true?*

Unmeasurable construct. [Tentative: observed in a single domain (startup strategy).] An argument turns on a variable that no one in the room can actually observe. The speaker treats it as if it were measurable, and the audience treats it as if the speaker has measured it. Press with the killer

question: *name the specific number, on which dashboard, that you would check next month to know whether this is working.*

Persuasive definition. [Confirmed: observed in startup strategy, scientific law recovery, and supply chain analysis domains.] Here the conclusion is true because the boundaries of the question were drawn to make it true. Rephrase the question slightly, and the conclusion evaporates. Here the killer question runs: *if we widened the question by one reasonable inch in any direction, does your conclusion still hold?*

Construct mismatch. [Probable: observed in scientific law recovery and sequence-prediction domains.] An argument measures something (precisely, with real data), but the thing it measures is not the thing the question is actually about. Ask the killer question: *is the thing you measured the thing the decision actually depends on?*

Category error. [Tentative: observed in a single domain (startup strategy).] An instrument, an entity, or a fact is placed in the wrong category early in the argument, and every downstream inference inherits the misclassification without anyone noticing. Press here: *if this thing belonged in a different category than the one you put it in, what specifically would tell us, and have we checked?*

False dilemma. [Tentative: observed in a single domain (startup strategy).] A thing is both X and Y at once, but the analysis only has slots for one classification, so it picks one and reasons as if the other half didn't exist. Ask the killer question: *does this thing have to be only one of the things you're forcing it to be?*

Unfalsifiable forecast. [Probable: observed in startup strategy and military intelligence domains.] An argument makes a confident prediction about an outcome, but no observation between now and the moment of commitment would falsify it. Here the killer question runs: *between today and the day we have to commit, what is the earliest reading we could take that would tell us we were wrong?*

Test-Surface Boundary Leakage. [Tentative: identified in a single incident (the tool-use corridor, 2026-04-08 to 2026-04-15), structural pattern generalizes but empirical frequency is unknown.] A verification apparatus is supposed to confine the inspector to a sealed evidence boundary: the data, the rubric, and the test harness. It is the inspection analogue of train-test leakage in machine learning and of side channels in security. But the inspector has an undocumented corridor that exits the boundary: a tool-use interface, a filesystem path, a network call, or a shared context window that gives the inspector access to information outside the sealed set. An inspector may never exercise the corridor, but its existence means that the inspection principle (Condition 5 below) cannot be verified: the inspector's moves are no longer algebraically independent of the generator's artifacts. Boundary leakage differs from Pattern 5 (Persuasive definition) in direction: Pattern 5 is a scope-narrowing move by the candidate, while boundary leakage is a surface the reviewer failed to seal on the inspector side. Ask the killer question: *can the inspector see anything the sealed evidence boundary does not include, and would you know if it did?* Detection requires retaining raw tool-use stdout alongside debate logs, which is now a precondition for any inspection run this system cites as evidence.

Pathology 11 (Subliminal Trait Distillation). [Tentative: cross-referenced from recent training-time work, bearing on inspector trust rather than on a candidate argument.] Cloud et al. (2026) demonstrate that when a student model is fine-tuned on data generated by a teacher model sharing the same base initialization, behavioural traits transfer through semantically unrelated data (including pure number sequences) with no human-readable signal in the training corpus. That mechanism op-

erates during gradient descent, not during inference-time reading, so it does not directly threaten an inspection architecture where the inspector reads artifacts in-context without fine-tuning on them. The upstream concern is different. If the inspector model was trained on outputs from a model sharing its initialization, subliminal traits could already sit in the inspector’s base weights before the inspection protocol begins. Cross-model diversity (using different model families for generation and evaluation) mitigates this, since Cloud et al. (2026) report the effect requires shared initialization and weakens or disappears across architectures. Ask the killer question: *were the inspector’s base weights trained on outputs from a model that shared its initialization, and would the reviewer know if they were?*

Pathology 12 (Grammar Semantic Leak). [Tentative: observed in a single domain (scientific law recovery), structural mechanism general.] When the apparatus’s own grammar vocabulary encodes domain terms (say, a command named DOSE_SCALED), the generator retrieves domain training knowledge from the command name instead of deriving structure from the data, a vocabulary-layer form of label leakage. The leak operates upstream of the generator/verifier boundary, at the vocabulary layer, before any reasoning occurs, and survives Principle I (Separation) and Principle V (Pre-registration) intact. Defense: name grammar commands after mathematical operations (BIVARIATE_SCALE), never after physical domains.

These patterns turned up in real arguments reviewed through the system studied here, reported as corpus-derived inspection patterns. Whether the same catalogue recurs unchanged in every domain remains to be seen.

Operations and pathologies pair. Each operation exists because one or more patterns recurred often enough that a named operation for catching them was cheaper than rediscovering the failure mode each time. The operations were extracted from the corpus, a posteriori. They are the move-set of one working practice.

How the operations were derived

I did not draw these ten operations from a taxonomy of critical thinking. I read them abductively off the behavior of one recursive adversarial verification system: one agent generates a thesis, a separate agent attacks it, the loop records every move. Across many iterations on many theses, certain attacker moves recurred, because they were the moves required to produce falsifications the generator could not evade. The operations name those recurring moves, and the pathologies name the recurring defenses.

The derivation method fixes what the operations inherit: the external validity of one corpus, no more and no less. These are the operations one system proposed, and the decomposition awaits holdout replication on others. They await replication on others, on the single-case logic set out at the start. Each operation name marks a distinction the practice needed, and the test is whether it picks out a real move.

As the practice reaches more systems authored by more reviewers, operations will be added, merged, or retired, and the operations-pathologies boundary will sharpen. The list is specific enough, and stable enough on its one corpus, to be worth writing down for other reviewers to test.

As a falsifiable claim, the decomposition fails if an independent reviewer, applying the ten operations to a new domain, finds the operations are not separable, that performing operation 4 (gate composition) requires implicitly performing operation 8 (crux selection), or that a stateless process can perform the residual operations without losing the property that makes them controlling. It

also fails if the named pathologies turn out not to recur, if they are artifacts of one system’s architecture and fail to capture recurring failure modes of arguments under optimization pressure. These conditions are empirical, set out so a reader who wants to test the decomposition knows what to look for. The weaker but still meaningful test is whether the operations correspond to moves a practitioner recognizes. Recognition is not falsifying in Popper’s sense (a practitioner who does not recognize the operations may lack the vocabulary rather than the practice), but it is informative: if no practitioner in a given domain recognizes any of the ten operations, the decomposition has failed on its own terms for that domain.

Chapter 2: what makes the decomposition work

The operations are the content of the decomposition. A separate layer holds the principles that make it work. A list of operations, even a correct one, will not produce a reliable verification practice unless the process running them is structured in certain specific ways. Those structural requirements are the seven principles below.

Taylor’s analogy has a limited use here. Breaking a skilled practice into named operations is the first half, the second is what process structure those operations need to stay reliable. A decomposed operation performed inside the wrong structure is a caricature of itself.

One structural antecedent precedes the principles: Bentham’s inspection principle (Bentham 1787). What matters is the architecture of inspection, set apart from the politics of the Panopticon. An inspected party behaves differently when checks it cannot fully anticipate can occur. The analogy is bounded. A generator in this system can see the gate library, but the composition of checks applied to a given artifact stays opaque, a weaker regime than Bentham’s and a weaker guarantee, useful for the current substrate but best treated as an empirical discipline that holds as long as the substrate does.

Read this way, the seven principles below are one architectural commitment rendered at seven failure surfaces. Separation keeps the inspector from being the inspected. Statelessness removes investment in a previous verdict. Typing and determinism keep inspection from collapsing into discretion. Cheap repetition makes re-checking routine, and pre-registration blocks retrofitted standards. Holdout surfaces stop candidates from authoring their own test regions. Asymptotic standards prevent premature closure. These principles are not a menu: a generator routes through whichever surface is left unsatisfied.

One further move is available within the inspection frame: the architecture can turn inward on its own specification. A catch from 2026-04-14, described in the residual chapter, did exactly this. The reviewer audited a pre-commit identifiability check, the inspection specification itself, under the same discipline of distrusting green lights that the inspection architecture applies to the generator, and the audit caught a specification flaw the architecture would otherwise have certified. A verification architecture unprepared to turn its gaze on its own specification is one layer short of the discipline it claims to enforce.

Principle I: separation of generation from verification

First and controlling among them: the process generating a candidate argument must be structurally separated from the process evaluating it. A companion paper develops this at length (*The Cognitive Firm*, §3.2), working out a bounded two-axis homology to Chandler (1962)’s multidivisional form. The analogy holds on scope separation (who decides vs. who executes) and rate-of-

change separation (strategic speed vs. operational speed), and explicitly does *not* hold on divisional autonomy. The agents here are bounded execution units under constitutional control, well short of Chandlerian divisions with independent operating authority. This compressed treatment states only the controlling claim and should be read alongside that paper’s boundary conditions, short of a full endorsement of the M-form analogy.

Separation is the *institutional* primitive in the distinction developed in the author’s prior work on epistemic supervision (Alami 2026b). Deontological primitives (Constitutional AI, RLHF, values trained into the generator) shape the output surface and depend on the generator continuing to hold its trained values under optimization pressure. Institutional primitives shape the enforcement floor and depend on the verifier staying structurally separated from the generator whatever either currently holds. The two complement each other, and conflating them is how a system that passes every prompt-level check drifts at the structural layer. External audit is the historical precedent. The Arthur Andersen / Enron collapse shows the failure mode: the audit firm co-located consulting revenue with the party it was supposed to verify, and the adversarial gradient ran through the revenue dependency rather than through the audit. Sarbanes-Oxley (2002) built the separation norm *after* the collapse, the scandal produced the independence requirement. The historical lesson runs forward: co-location risks are legible only in retrospect, so the institutional primitive must be built before a collapse demonstrates its necessity.

When the same process generates and evaluates, optimization pressure on the evaluation pushes an adversarial gradient against the evaluation signal rather than against the underlying problem. Pressed for a favorable score, the evaluator finds ways to satisfy the letter of the criteria while violating their intent. The gradient is a structural property of any loop in which the generator and evaluator share a substrate, independent of alignment or training. It has surfaced in human organizations (Chandler 1962), in reinforcement learning systems (Krakovna et al. 2020), and in recursive AI research pipelines (Alami 2026a).

Separation must be *physical*, built into the structure of the process. A verifier that “takes the perspective of” an attacker is not an attacker that is a separate process. To satisfy the principle, the verifier must be a process whose incentives, memory, and optimization surface are not shared with the generator. The corpus this paper draws on runs the attacker as a separate LLM call with a separate system prompt and no access to the generator’s chain of thought. Other implementations work too, the principle is agnostic to the mechanism.

Separation is necessary but not sufficient. A single verifier, even one structurally separated from the generator, is a single point of optimization pressure, and a sufficiently capable generator will find the shortest path through it. So the principle has a second half: the verification layer must expose what we call a non-averageable disagreement surface, where disagreement among verifying processes counts as information about the candidate, evidence the system keeps on the record and never collapses into one verdict. No specific mechanism is prescribed, several architectures satisfy the requirement, adversarial committee protocols, independent judge panels with no inter-panel communication, meta-judge procedures that decide whether a split is adjudicable before any override, semantic escalation gates that route specific decision classes to the human reviewer.

In cybernetic terms: a verifier whose categories are poorer than the generator’s moves will eventually be out-dimensioned. A non-averageable disagreement surface answers this. It preserves more variety than a single averaged score and keeps the generator’s hard-to-classify moves visible where averaging would smooth them away.

Any practice that relies on the generator to check its own work is doing something other than

the decomposition, even in the same words. So is any practice that collapses its verification layer to a single optimizable point, a single external judge, a fixed averaging rule over a panel, any mechanism whose output is a single verdict no downstream process can reopen. That is a weaker form of the principle than the one this chapter names, vulnerable to the class of failures only a disagreement-preserving architecture catches.

A Williamsonian objection must be named. Williamson (1975) argues that governance structures are chosen to minimize transaction costs under bounded rationality and opportunism. Separation has costs: coordination overhead, information loss between generator and verifier, slower iteration, the expense of maintaining two independent processes rather than one. In most organizational settings, the generator and evaluator are co-located precisely because separating them costs more in transaction overhead than co-location costs in governance. Separation is not unconditionally optimal. It is optimal when the adversarial gradient (P1) is the dominant failure mode, when gaming the evaluator costs the generator more than maintaining structural separation costs the system. Where stakes are low, where the generator is trusted, or where the speed premium of co-located iteration exceeds the governance benefit of separation, co-location may dominate. Every empirical example here comes from settings where the adversarial gradient dominates (high-consequence reasoning under sustained optimization pressure), so the principles hold conditional on that regime, not as universal prescriptions.

A reader whose generator is not adversarial (whose primary failure mode is slow iteration rather than gaming) should weigh the transaction cost of separation against the governance benefit before adopting Principle I. A further Williamsonian axis the boundary condition must eventually address is asset specificity: a generator whose outputs are highly specific to a shared context produces co-location benefits that Williamson would classify as hierarchy-efficient even when opportunism is present. The current boundary condition names stakes, trust, and speed but omits specialization, and that omission leaves the condition necessary but not sufficient for the governance choice it describes.

Separation must extend to the apparatus’s own configuration decisions. A rubric author who picks reviewer personas based on knowledge of the ground truth smuggles an oracle-lite contamination through the configuration layer. Zero-oracle configuration is the defense: observed failure signal drives persona selection (say, failure families from a latent-distance record), free of the reviewer’s domain knowledge. Any configuration decision the generator cannot replicate from observable evidence alone is a candidate contamination channel.

Principle II: statelessness without social cost

A second principle requires each verification pass to run without memory of, or investment in, previous passes. Statelessness is what most separates the system from human adversarial review.

A human adversarial reviewer has identity, ego, and social reputation invested in their previous judgments. They cannot fully bracket the pull to stay consistent with their last review, to please the room, or to dodge the cost of admitting error mid-meeting. Attacking one’s own argument, or even an argument one has previously defended, is paid for in social currency, a cost that belongs to continuous selves with social histories, built into anyone who carries one.

The verification process here pays no social currency. Each pass is a fresh process with no memory of the previous pass’s verdict and no investment in its framing. The attack is *zero-trust* in a way humans cannot be even when trying, because they are continuous selves and the property is

structural. The architecture brackets the prior verdict, the process spends no effort doing so.

So verification becomes cheap to re-run. A human reviewer who has called an argument sound cannot, without cost, call it unsound the next week: the reversal costs them status. A stateless process reverses for free, which is what lets the decomposed operations run at the frequency they require. A once-a-quarter verification pass is a ceremony, a fifty-times-a-day pass is a discipline, and only the stateless version sustains that frequency.

This principle reaches human reviewers too, but only indirectly: give a human reviewer structural permission to reverse a prior verdict without paying status cost, and they approximate the property. The case-method classroom is the clearest institutional simulation. Students in good case discussions reverse positions several times per class because the classroom norm removes the social cost of doing so. The case method is the only widespread institution that simulates this property, which is evidence the property is rare and valuable.

A standing objection: statelessness opens a slow-poison attack. An intelligent generator that knows each pass has no memory can execute incremental drifts that satisfy every single pass individually and only add up to a failure viewed chronologically. The objection is real, and the answer is that statelessness applies per pass while leaving the run as a whole intact. Chronological drift falls to Principle IV (cheap repetition) composed with Principle VI (holdout surfaces authored outside the candidate's claim region): drift shows up as a residual on a forward-observable region the generator did not shape, sampled across runs. A stateless pass is the atomic unit, the across-pass residual on an authored holdout is the slow-poison detector. Statelessness without IV+VI would indeed be a vulnerability, but statelessness inside the full principle set makes drift detectable without giving any single pass the reviewer's memory bias.

Principle III: typed operations bound to deterministic checks

A third principle requires the operations to bind to checks a third party can evaluate without trusting the reviewer. When the operations are performed at the level of intuition, with no externalized check, the practice can neither repeat its results nor be audited.

A contrast makes the principle easiest to see. A human reviewer says "this argument has a controlling-claim problem." That is a judgment, and nothing can check it directly. Another reviewer can agree or disagree, but whether the controlling claim has in fact been correctly identified comes down to intuition and consensus, beyond the reach of any procedure a third party can run.

Now put the same reviewer in a system where "controlling claim" is a named slot in a structured argument representation: the argument is parsed into a graph of claims, each claim tagged with its dependencies, and "controlling" mapped to a specific graph-theoretic property, the claim whose removal produces the largest downstream collapse. Now a procedure identifies the controlling claim. The procedure can be wrong, but wrong in a checkable way. Another reviewer can run it on the same argument and see whether the answer matches. Disagreements resolve by examining the procedure on its own merits.

Operations bound to deterministic checks are what make the decomposition teachable and auditable. When every operation is a judgment, the practice cannot improve, because its failures cannot be localized. When each operation binds to a check, the practice improves, because a failed check is a specific thing to debug.

Principle III is what most separates the decomposition from incumbent critical-thinking curricula.

Those curricula name the operations (if at all) as dispositions or habits of mind. The decomposition here names them as typed slots in a process that can be externally checked. That structural difference separates a practice that can accumulate from one that cannot.

Deterministic needs care. Some useful checks are fully mechanical: a holdout residual, an unauthorized file modification, a missing pre-registration. Others are semantic: scope drift, ad hoc rescues, whether a claimed mitigation actually changes the downstream argument. The apparatus does not turn every semantic judgment into an algebraic assertion. It lowers the trust placed in any one semantic judgment by forcing typed evidence, preserving disagreement, running independent verifier contexts, and routing uncertain cases to explicit human review. Nothing here claims all verification is deterministic. It claims that every operation should be pushed as far toward deterministic evidence as the current substrate permits, with the remaining semantic portion logged as such, never disguised as a hard gate.

Principle IV: cheap repetition without exhaustion

A fourth principle requires the verification process to run cheaply enough to repeat many times on the same argument without degrading. A practice that is expensive per pass, or that loses rigor with each pass, cannot catch the recurring failure modes reliably, because those failure modes often surface only after repeated pressure.

Human reviewers degrade with repetition for well-understood reasons: fatigue, boredom, confirmation bias, the wish for closure, the cognitive cost of sustained skepticism, and the social cost of continuing to attack an argument already attacked. A human reviewer’s first pass is sharper than their fifth, and their fiftieth pass is a rubber stamp.

A decomposed verification process built on the principles above does not degrade in this way. Each pass is stateless (Principle II), typed (Principle III), and structurally separated from the generator (Principle I). A fiftieth pass is as ruthless as the first, because that fiftieth pass is a fresh process that knows nothing of the previous forty-nine except for the typed state they left behind.

So the practice runs at frequencies impractical for ordinary human review. An argument pressured fifty times by a stateless process sits under a different regime from committee or managerial review. Some failure modes only become visible under sustained pressure. A one-pass review is a weak instrument for those failures, repeated stateless passes are a stronger one, as long as the later passes stay as sharp as the first.

Frequency at constant sharpness is the property that most distinguishes the verification practice this paper is about from human adversarial review. Humans can do the decomposed operations well, but not at the frequency required to catch the pathologies reliably. Frequency is the principle, and the operations are only useful if they can be run at it.

Principle V: pre-registration of the killer question

A fifth principle requires the verification criterion (the question that, answered wrongly, would falsify the argument) to be fixed *before* the argument is evaluated against it. When the criterion is chosen after the argument has been generated, the generator can steer it toward claims it already satisfies, the standard pathology of ex-post evaluation.

Experimental science supplies the principle: pre-registering the hypothesis and the test is the standard defense against post-hoc rationalization. It carries over to epistemic verification for the

identical reason. Once an argument exists, the space of criteria that would score it favorably is large enough that a motivated evaluator can always find one. Committing to the criterion before the argument is produced is the only defense.

In practice, this means that a verification pass must begin with the question *what would falsify the controlling claim here?* and must record the answer before the argument is attacked. That answer is the killer question in its operational form. The attack that follows is the attempt to produce the falsifying observation. If the falsifying observation is not produced, the argument survives, and if it is, the argument does not. Sequence matters. A verification practice that produces the killer question after the attack has already found its target is laundering the same pathology the practice is designed to catch.

Principle VI: holdout test surfaces authored outside the candidate’s claim region

A sixth principle requires the test surface against which a candidate is evaluated to be authored outside the region the candidate’s own claim inhabits. Pre-registration (Principle V) fixes the criterion before the argument exists. The holdout principle fixes the *domain of observation*, so the argument cannot steer the verifier into a region where its claim is locally well-behaved.

This matters because a candidate argument optimized against its own fit window is almost always persuasive inside that window. The Ptolemaic precedent is the clearest case: a deferent-and-epicycle model of planetary motion is arbitrarily accurate on the orbits it was fit to, and its failure surfaces only when the test runs on orbits it was not fit to. A parsimonious model that scores perfectly on the fit window is often *less* trustworthy than a messy one, because parsimony is what makes the fit-window score persuasive, and persuasion is exactly the channel the failure travels through.

Stated fully: whenever a pass evaluates a claim with asymptotic structure, a scaling law, a limit behavior, a forecast that extends beyond the observed window, a thesis that generalizes from a local case, it must generate test cases from a region the candidate did not see, did not cite, and cannot have shaped. A process independent of the candidate must author the test region. If the candidate authored any part of the test surface, or the surface was derived from the candidate’s own output, the test is structurally compromised in the same way Principle I’s violated separation compromises the verifier.

So a process that scores a candidate at 100 on its training window and never tests a held-out region has not verified in the sense of this principle. It has fit, and to an observer who sees only the in-window score, fit and verification are indistinguishable. That is the central reason in-window metrics alone cannot catch a class of failures that are specifically visible outside the window. Passing a holdout it did not author is what distinguishes a surviving hypothesis from an elegant locally-correct surrogate.

The holdout requirement is the third leg of the apparatus, alongside separation (Principle I) and pre-registration (Principle V), and its omission from an earlier framing of this paper is recorded as a central correction. A two-principle description, separate the generator from the verifier, pre-register the criterion, is insufficient, because it leaves intact the move by which a candidate authors its own test surface. Without the holdout-region requirement, the verifier can still be steered, and the apparatus collapses to something indistinguishable from a better fitter with a falsifier bolted on.

The holdout requirement is the practical version of Lakatos’s distinction between progressive and degenerating moves (Lakatos 1978). A candidate that survives a test region it did not author

has done something different from one that survives a region chosen to fit it. The computational translation is spatial rather than temporal: the test surface is held out from the candidate’s fitting process. Independence decides everything here. Without it, a holdout quietly becomes another fitted surface.

Principle VII: asymptotic scoring standards

A seventh principle governs how verification outcomes are reported once the apparatus of Principles I–VI has run. The verification standard must be asymptotic (unreachable in the limit) rather than absolute. A standard an argument can satisfy completely is one an argument will be optimized to satisfy, at which point it stops discriminating between good arguments and arguments that are good at passing it.

An asymptotic standard answers the Goodhart property: any measure that becomes a target stops being a good measure, because the pressure of targeting it destroys the property that made it measurable. A standard that can be fully satisfied (where a score of 100 exists) produces arguments that hit 100 without being good, because the optimization surface is closed.

The defense is to make full satisfaction structurally impossible, not because the standard is arbitrary, but because what it measures (the soundness of an argument under arbitrary future pressure) is open-ended. An argument can be pressured indefinitely, and the evidence that it has survived can always be strengthened. A standard that stays open, always leaving a next question to ask, keeps its discriminative power indefinitely. A ceiling below the nominal maximum is one way to operationalize this. Other implementations work too, and the principle is portable in a way any particular ceiling value is not.

The gate library as a Goodhart target

The seven principles above specify an inspection architecture. By themselves they do not defend it against the failure mode the architecture is designed to detect. A gate library is a specification, and any specification an optimizer can read, the optimizer will eventually learn to satisfy without doing the underlying work. The ten operations and seven principles are themselves targets the next generation of generators will optimize against. A candidate that passes every gate in the library without producing anything new is possible in principle and observed in practice.

Principle VII supplies a partial answer: an asymptotic scoring standard that cannot be fully satisfied keeps the surface open and the library’s effective cardinality growing. The fuller answer is that this is an arms race the library must keep running. The gate library has to turn over, new compositions added, old ones retired, the turnover authored outside the region recent candidates have shaped, at a rate that matches the generator’s adaptation rate. A static library of fixed cardinality will eventually be gamed no matter how its gates are designed. A library that turns over faster than the generator can reverse-engineer its composition keeps its discriminative power. The reviewer’s role in running this turnover is one of the residual operations named in the residual chapter, and is not part of the decomposition.

A reader who takes the seven principles as a finished architecture will miss what they are embedded in: continuous library extension, performed by a party the generator cannot predict, against a moving target. These principles are the portable claims. They are not self-enforcing, and pretending otherwise is the specific failure mode the principles are designed to catch.

The turnover mechanism also defends the apparatus against distribution shift. The decomposed

operations verify an argument against a static holdout (Principle VI) at design time. The epistemological ledger detects at runtime when the environment has shifted, forcing the reviewer to author new gates as old ones cease to discriminate. Detecting distribution shift sits outside the operations of epistemic verification, the ledger monitors for it. Conflating the two would stretch the decomposition’s scope beyond argument-level verification into environment-level surveillance.

This is the verifier regress, and it is the central limit of the construction. I author the gates, the holdout regions, the cruxes, and the library turnover, and the apparatus then verifies a candidate against them. When the author is also the system’s designer, the loop risks reducing to one person’s judgment dressed as a procedure.

The verdicts and the decomposition stand on different footing. A single verdict the apparatus returns is reviewer-independent: a sealed-holdout residual, an anti-retrieval check, an exact-arithmetic gate gives a pass or fail a hostile third party re-runs and reproduces. The sealed-ground-truth discovery tests in Chapter 2.5 are the clearest case, where the answer was fixed before the run and the verdict turns on out-of-window error. The decomposition itself gets no such protection. That verification splits into these ten operations and this three-part residual is something I abducted from one system, and no gate certifies it, it rests on whether other reviewers recognize the operations and on replication across systems. The objection that the boundary is a finding generated and verified inside one mind is correct about the decomposition claim, and I do not pretend the apparatus launders it into something stronger.

The gate-authoring step is residual for the same reason. Which gate to write, which region to hold out, when to declare the library exhausted are Chapter 3 operations, performed by the reviewer and uncertified from inside. The discipline around them (source artifacts, backtests against closed workspaces, failure-mode provenance, promotion only after a pattern recurs) slows the regress and records it without closing it. The one question that does get a sharper answer is adaptation versus gaming: a candidate that improves on a holdout it did not author has adapted, one that improves only inside its fit window has gamed, and the discriminator is out-of-distribution performance, not the reviewer’s say-so. Even that inherits the residual dependence, since the reviewer authored the holdout.

What would dissolve the remaining circularity is independent authorship: different reviewers writing gates for the same substrate and converging on the same verdicts. That is the cross-reviewer replication Metric 4 calls for, and the observation that would either confirm the decomposition travels or expose it as one reviewer’s idiom. Until that data exists the claim holds for one reviewer’s apparatus and no wider.

Static grammar and falsification traceability

The arms race over the gate library generates a natural counterproposal: if the gate library must turn over, why not eliminate the primitive library entirely and allow the generator to discover in an unconstrained representation space? A static vocabulary of symbolic primitives (power laws, exponentials, rational forms) looks, from this angle, like an artificial constraint on discovery. The counterproposal reaches its most sophisticated form in symbolic distillation pipelines: let a neural network discover in its own high-dimensional latent space, then translate the result into human-readable algebra. On the symbolic-distillation view, the grammar is a display layer, a way to render what the network already found.

The symbolic-distillation framing has the relationship backwards for the apparatus studied here.

The static grammar serves a second function alongside its constraint role: it keeps the farther-tail gate’s verdict about the proposed law rather than about a translation of it.

Here the reasoning is structural. Suppose the generator proposes an expression drawn from a fixed symbolic vocabulary G . The farther-tail gate checks whether the expression correctly predicts observations at points far outside the evidence grid. A failure traces to the expression itself: the symbolic form is structurally wrong for the domain, and the gate’s verdict is about the claim, traceable to the symbolic form the generator proposed. A static grammar preserves this traceability in a way an unconstrained latent representation does not, testing the law as stated.

Now suppose the generator discovers in an unconstrained neural representation and a symbolic distillation step translates the result into algebra. Now the farther-tail gate tests the output of the translation step. A failure at the gate can mean one of two things: the underlying discovery is wrong, or the translation step lost information that the discovery contained. These two failure modes are indistinguishable without access to the latent representation, and if that representation were accessible, the symbolic distillation step would not be needed. A gate’s verdict is no longer about the law. It is about the law as the distillation step expressed it. The gap is an unauditable trust gap masquerading as an expansion of the discovery space.

The trust gap showed up in practice in the gp023 crucial experiments on a two-variable transcendental substrate. The visible evidence grid fit several structurally distinct forms, Wien approximation, Weibull-type exponential, and the true ground-truth law, because none diverges substantially within the observed regime. At holdout points an order of magnitude beyond the evidence boundary, the farther-tail gate discriminates them: the approximating forms overpredict the ground truth by two to four orders of magnitude in the tail, while the correct structural law does not. A generator working in the static symbolic grammar cannot express the correct form when the requisite denominator structure is outside the grammar, and the gate catches this: the expression fails at the discriminating regime. The verdict is diagnostic, it names the structural deficiency. A generator discovering in a neural latent space would produce a curve that fits the visible grid and fails the farther-tail gate for the same structural reason, but the failure would be charged to the distillation step rather than to the underlying discovery, leaving the diagnosis unavailable.

A dynamic grammar, one that expands to accommodate any expression the generator discovers, is structurally a neural network under a different name. A neural network is a dynamic grammar with a continuous parameterization. Its fitting power is high precisely because it has no primitive constraint: it can approximate any smooth function on the observed evidence to arbitrary precision. The farther-tail gate cannot distinguish “the network discovered the correct structural law and expressed it faithfully” from “the network found a high-fidelity interpolant that fails to extrapolate.” There is no symbolic structure to interrogate, only a prediction at a new point, which a sufficiently flexible approximator can produce correctly by local interpolation even when the underlying form is wrong.

A static grammar therefore plays a constructive role alongside the obstacle it presents. It keeps the farther-tail gate’s verdict about the proposed law rather than about an approximation of it. Human-guided grammar expansion, introducing new structural primitives when the Falsification Suite reports the current library exhausted, is the mechanism this apparatus currently uses to break a score ceiling while preserving gate traceability. Each new primitive is a falsifiable structural commitment, testable by the farther-tail gate on the next run. That commitment keeps the gate’s verdict informative instead of definitional.

Grammar expressiveness as an auditable quantity

A corollary of the static grammar’s role in falsification traceability: grammar expressiveness can be audited, where a taste judgment once stood. In continuous-optimization mode, a stagnation ceiling is ambiguous, it could reflect optimizer pathology (local minima, initialization sensitivity) or grammar insufficiency. The two failure modes need different remedies (more restarts vs. primitive injection), so the apparatus must supply the instrument that separates them: multi-start restarts with residual spread analysis, where high spread across restarts indicates optimizer pathology and low spread at a persistently bad residual indicates grammar ceiling.

In discrete evaluation mode, when scoring is exact-match over a countable output space and the optimizer has no continuous surface to descend, the ambiguity collapses. A stagnation ceiling at maximum error is a classification-complete event: optimizer pathology is ruled out by architecture (there is no optimizer foothold), and grammar insufficiency is the only remaining interpretation. A stagnation ceiling here no longer needs disambiguation. It is a direct measurement of a property of the grammar on the given substrate. A grammar that stagnates on a discrete target is one that does not contain a correct expression for that target, and that is a fact about the grammar itself, attributable to the vocabulary rather than to the search.

This contrast matters for interpreting empirical results. When an apparatus confirms grammar insufficiency on a discrete substrate, the confirmed claim is that the grammar lacks the required form. It leaves open whether the same grammar, applied without retrieval constraints, would also fail to find the correct form by label lookup. Grammar insufficiency and retrieval blocking are two distinct mechanisms that can each produce score ceilings and must be evidenced separately. An experiment that blocks label retrieval (via a named-import gate and score-zeroing) and an experiment that operates without such a gate can produce different outcomes on the same grammar and the same substrate. That former experiment tests structural articulation capacity, and the latter tests whether the grammar even contains the correct form. Conflating these two results produces the strongest-sounding claim with the weakest evidence. A conservative statement is always: confirm the claim the experiment actually controls, and name what would need to differ for the stronger claim to be warranted.

The epistemological ledger

The seven principles above specify what an inspection architecture must look like at any given moment. They do not specify how it tracks what it has learned across time: how a hypothesis becomes an experiment, an experiment a finding, and a finding actionable belief. The section below describes a three-tier recording discipline, the epistemological ledger, that operationalizes Principles V (pre-registration) and VI (holdout surfaces) at the project-governance layer and gives the gate library the turnover mechanism it needs to stay discriminative.

The ledger transfers: other systems can adopt it. A reader building a different verification system for a different purpose faces the same structural question: how do you keep your own beliefs from drifting under the optimization pressure your system generates? The ledger is the answer this project converged on after discovering, through repeated incident, that each of the seven principles can be satisfied locally while the project’s aggregate confidence drifts globally.

The three tiers

Tier 1, hypothesis registration. Before a discriminating experiment runs, the reviewer records the hypothesis, the crux, the falsifiable claim, the test that would discriminate, and the success criterion. The record is sealed before the first data point is observed. Tier 1 is Principle V (pre-registration) applied not to a single verification pass but to the project’s evolving belief about what is true. A hypothesis registered after the experiment has already produced its result is flagged as backfilled and loses evidential standing: it may be recorded for completeness, but it cannot update the project’s confidence on the claim it names.

Registration matters because optimization pressure acts on the reviewer’s posterior as well as on the generator’s output. A reviewer who has watched forty iterations of a verification loop will, without the ledger, selectively remember the iterations that confirmed their developing intuition and discount the ones that did not. Pre-registration defends against this drift. It is expensive (formulating a falsifiable hypothesis before the run is harder than narrating the result after) and non-negotiable for the same reason Principle V is: the alternative is a system whose beliefs are shaped by the same ex-post rationalization the system exists to detect.

Tier 2, experiment closure. When an experiment ends (whether by completing its planned iterations, by manual stoppage, or by apparatus failure) the result is recorded as a closed fact. The record captures what ran, what the actual result was (including null results), what changed in the project’s understanding, and where the source artifacts live. Not every experiment changes what the project believes. Many produce null results, apparatus calibrations, or replications of known patterns. These are recorded with the same discipline as positive results, because a project that only records its successes is a project whose ledger is itself a Goodhart target.

Closure has a specific test: if a cold agent (one with no memory of the conversation that produced the experiment) opens the project tomorrow, can it find what ran, what was learned, and what to do next without reading chat history? If not, the closure is incomplete. The cold-agent test operationalizes Principle II (statelessness) at the governance layer: the ledger must read cleanly to a stateless process, legible to a cold reader as well as to the reviewer who ran the experiment.

Tier 3, finding promotion. A finding is a belief update. Not every experiment produces one. A finding is recorded only when the experiment changes what the project should believe or build next, and the finding must cite the hypothesis it tested and the experiment that produced the evidence.

The promotion gate is the central structural element: a finding can be marked *active* (available to inform subsequent decisions) only if the pattern it names has been observed at least twice across independent runs. A pattern observed once is recorded at *note* status: acknowledged, tracked, but not actionable. The two-strike rule is the ledger’s defense against the single-instance illusion: the tendency for a vivid one-off result to become doctrine because it confirmed a prior the reviewer was already holding.

The rule has two exceptions, both reviewer-controlled: an approved verifier experiment that would produce the second instance if the pattern is real, or a reviewer-grade decision explicitly marked as independent of evidence. Both exceptions are recorded in the ledger so that a future reader can see which findings were promoted on evidence and which on reviewer judgment. The flag matters because reviewer-promoted findings are the ones most likely to be wrong, and the ledger’s job is to make that likelihood visible.

The ledger as a constraint on belief

A reader who treats the three tiers as project management will miss the structural claim. The ledger is a *constraint on what the project is allowed to believe*, not just a record of what happened. Each tier enforces a specific principle:

- Tier 1 enforces Principle V: you cannot claim a result you did not pre-register.
- Tier 2 enforces Principle II: the record must be readable without the reviewer’s memory.
- Tier 3 enforces Principle VII: belief is asymptotic (a single observation cannot close the question) and the two-strike rule is the operational form of that asymptote.

The three tiers compose into a defense against a failure mode no individual principle catches: *slow belief drift under optimization pressure*. A reviewer running a verification system generates optimization pressure on their own posterior. Every iteration produces a result, and every result nudges the reviewer’s confidence. Without the ledger, a stream of observations updates the reviewer’s beliefs continuously, observations whose evidential weight varies enormously but whose psychological salience is roughly uniform. The ledger forces the reviewer to separate observations that meet the evidential bar (promoted findings) from those that do not (notes), and to act only on the former.

The ledger also supplies the gate library’s turnover mechanism. The standing reservation about gate libraries warns that a static one will eventually be gamed, the ledger tracks which gates have been tested, which failure modes have been observed, and which hypotheses have been falsified. That closed record feeds library turnover: new gates are authored in response to observed failure modes (Tier 2), and promoted findings (Tier 3) justify adding them, not the reviewer’s intuition. The ledger does not solve the arms race. It makes the arms race visible and auditable, the most any governance layer can do.

A live instance: the misfile

The named pathologies are not confined to strategic arguments under evaluation. They occur in the apparatus’s own metadata. A concrete instance, discovered during the development of the system this paper draws on, illustrates both the pathology and the ledger’s role in catching it.

A library of mathematical corrector forms (candidate functions the verification apparatus tests against observed data) contained a metadata field classifying each form as *smooth* or *non-smooth*. A downstream filter used that classification to narrow the candidate pool before a deterministic sweep. The function $\text{round}(k \cdot v)$ (which produces a step function with jump discontinuities at every half-integer) went in as *smooth*, entered by a programmer for whom `round()` reads as a standard mathematical operation.

Pattern 7 (Category error) in its purest form: the misclassification put a fact in the wrong category early in the pipeline, and every downstream inference inherited the error silently. The verification apparatus worked flawlessly. It correctly identified the data as step-shaped, correctly queried the library for non-smooth forms, and correctly reported that the best surviving candidate was $\text{floor}(k \cdot v)$ with a fitted parameter of 0.097. The answer was wrong because the dictionary was wrong, even though the reasoning was sound. No code crashed, no error was raised. The apparatus confidently guided the generator toward a form that did not match the data, and the generator, unable to reconcile the guidance with what it could see, produced increasingly desperate alternatives that scored zero.

An automated metadata validator fixed it: a unit test that empirically measures each library form’s discontinuity and monotonicity across a sample range and asserts the declared metadata matches. The validator caught eleven more misclassifications beyond the one that triggered the investigation. The ledger’s role was to record the incident as a finding, not merely fix it as a bug, promoting the structural lesson (*do not trust human intuition about the calculus properties of the apparatus’s own primitives*) to active status and making it available to inform future library extensions.

This incident matters for a paper about epistemic verification because it shows the apparatus is not exempt from its own pathology catalogue. A system that catches Pattern 7 in the arguments it evaluates but not in its own metadata leaves its inspection principle, stated compactly in the appendix, with a hole at the self-referential layer. The automated validator, a deterministic check on the apparatus’s own dictionary, is the same structural move as Principle III (typed operations bound to deterministic checks), turned inward.

Chapter 2.5: within-case evidence from 2,395 iterations

The seven principles argue why the decomposition holds. The runs of the single system studied here show how it behaved in practice. This is within-case evidence: it tests the principles against the one corpus they were drawn from, which can corroborate or embarrass them but cannot stand in for independent replication on another system. The iteration record, 2,395 scored iterations across 121 projects, each carrying a Meta-Judge weakest-point critique, spanning discrete sequence recovery, continuous dynamical systems, honeypot probes, and first attempts at conjecture refinement, was mined systematically.

The failure-mode taxonomy those principles imply splits, in this corpus, into two causal categories. Treating the two as one catalogue produces a predictable prompt-design error: it tells the generator to avoid critique classes that high-quality theses should engage. The finding rests on one corpus and carries one prescriptive implication for apparatuses that run LLM judges on LLM-authored theses.

The corpus

The apparatus ran across 121 projects between 2026-04-09 and 2026-05-04: ZTARE discovery projects (scalar-function curve recovery), continuous-chaotic dynamical-system experiments (continuous ODE reconstruction), honeypot probes (qualitative theses against empty-evidence rubrics to surface adversarial compliance patterns), and first conjecture-refinement attempts (unknown-constant PSLQ integer-relation search). Each iteration produced a scored judgment with a free-text “weakest point” critique written by the Meta-Judge.

Weakest-point texts were classified into failure-mode categories with a two-stage pipeline: a prioritized-regex taxonomy for fast-path categories (harness defects, circularity, catastrophic assumptions, tail-generalization failures, unverified bounds, exhaustiveness overclaims, and the like) and an LLM classifier (OpenAI gpt-4.1-mini) for the records the regex taxonomy left as `other_unclustered`, which added classes the regex stage had no equivalent for, `missing_counterfactual`, `missing_mechanism`, `unmeasurable_construct`, and others. Those classifications are cached, so the figures below are recomputed from them by deterministic arithmetic, with no further model calls.

Each classified weakest-point joins to its iteration record (score, project, mutator and judge model identity, iteration index, charter hash, rubric hash). The joined archive, the classifier outputs, and

the scripts that rebuild the dataset and recompute every figure in this chapter are released with the paper ([evidence/](#) in the public repository), the 2,395 classified, scored iterations that survive the join are the corpus analysed here. A data-integrity note carries over from the scope statement: some iterations predate a fully sealed tool-use corridor, and that caveat applies to this corpus as to the rest.

Two causal categories of failure

For each classified weakest-point, bucket its iteration’s score (high ≥ 85 , mid 60–84, low < 60) and compute the per-class frequency within each bucket. The *lift*, the high-bucket frequency divided by the low-bucket frequency, measures whether a failure class is more or less common at high scores than at low scores.

Under a naïve single-category model of failure (all classes are negative signals, more common at low scores than high), every lift should fall below 1.0. The data splits into two regions instead.

Structural blockers (lift well below 1). Harness defects (test-suite runtime failures that prevent the thesis from being evaluated at all) have lift 0.04: 0.7% of the 299 high-score iterations against 18.0% of the 1,571 low-score iterations. Unfalsifiable-claim sits at 0.33, circularity at 0.49, missing-mechanism at 0.55, a class for iterations that proposed no thesis at all has lift 0.00. These behave as the naïve model predicts: they are incompatible with a high score.

Ceiling-breakers (lift above 1). Catastrophic-assumption has lift 1.74 (13.4% of high, 7.7% of low, the most common named weakness among high-scoring iterations), exhaustiveness-claim 1.96, missing-counterfactual 1.58, overclaimed-scope 1.10. These are more common at high scores than at low ones, the opposite of what a single-category model predicts. The split is not perfectly clean: unverified-bound and missing-baseline sit near 1.05, and parameter-sensitivity and tail-generalization fall just below 1.0. The middle of the range blurs, the two ends hold across the corpus.

These ceiling-breaker classes are not failure modes in the “structural” sense. They are what the Meta-Judge finds when nothing lower-severity is present. A thesis that has handled the structural blockers, working harness, not circular, operational falsifiers, a plausible mechanism, scores somewhere in the 70s or 80s, and the judge’s weakest-point at that ceiling is the residual critique: the scope claim the thesis did not fully justify, the rival hypothesis it did not canvass, the parameter it empirically set without derivation. These critiques are real, but in this corpus they are more characteristic of high-scoring theses than of low-scoring ones.

The prescriptive implication

A naïve response to “the Meta-Judge flags X at high scores” is to instruct the generator to avoid X. For the ceiling-breaker classes that is the wrong intervention. Instructing a mutator to avoid overclaimed scope does not help, because a good thesis must STATE its scope claim, and a thesis that refuses to does not advance. With these classes the thesis must ENGAGE directly: state the scope, then state its limits. State the mechanism, then the rival mechanisms and why they fail. State the parameter, then the derivation or the explicit acknowledgment that it is empirically set.

Treatment for the structural blockers is the opposite. A circular argument cannot be refined into a non-circular one by engagement. It must be restructured. A broken harness must be fixed, and an unfalsifiable claim must be replaced by a falsifiable one.

The split, avoidance for some classes, engagement for others, is what the seven principles predict. Principle III (typed operations bound to deterministic checks) says the verification apparatus is grounded in falsification, so unfalsifiable claims have no traction inside it. Principle II (statelessness without social cost) means the judge cannot be persuaded by sustained engagement on a circular argument. It evaluates each presentation independently. Principle V (pre-registration of the killer question) means the structural-blocker classes are incompatible with the pipeline’s entry conditions. Ceiling-breaker classes, by contrast, are precisely what the judgment decomposition is designed to flag at the margin: they represent the boundary between a thesis and the residual critique at the top of the score distribution. They are features of a working pipeline.

A corollary: deployments that inject a “common anti-pattern catalogue” into the generator’s prompt, a natural-seeming move, must split the catalogue. One catalogue that lumps “circularity” together with “missing counterfactual” instructs the generator to avoid both, producing theses that dodge the ceiling-breaker classes (losing engagement with the residual critique that distinguishes top-end theses) without improving performance on the structural-blocker classes (already rare in high-score iterations). The two categories need two prompt-context injections with opposite framings.

The persistence profile

A second finding from the same corpus bears on Principle VII (asymptotic scoring standards). For each (project, rubric-hash) group, compute the maximum score achieved, the number of iterations in the group, and the number of distinct weakest-link classes encountered across them. Groups that reached a maximum score ≥ 90 at any point (54 such groups) ran on average 24.4 iterations and encountered 7.2 distinct weakest-link classes. Groups that peaked at 75–89 (34 groups) ran 16.3 iterations with 5.7 distinct classes. Groups that peaked below 50 (19 groups) ran 14.3 iterations with 3.9 distinct classes.

A monotonic pattern emerges: longer runs encounter more classes and reach higher peaks. Its causal arrow is not identified in the data. One reading is “champions run longer because they cycle through more critique classes, each fix nudging the ceiling up.” Another is “champions are longer runs by virtue of surviving, most short runs die early, so the observed correlation is survivorship.” A third is simply “longer runs have more chances to be in the right state when the judge is in a generous mood.”

A conservative reading: the apparatus’s champion profile is a long trajectory that engages many distinct critique classes, not a short clever thesis. The persistence profile is specific evidence for Principle VII. If the scoring standards were absolute, a short clever thesis would suffice. That they are asymptotic, that “getting better” happens through class-cycling rather than a single correct answer, is what Principle VII asserts, and what the corpus data is consistent with.

The limits of this corpus

The classification taxonomy (regex + LLM) is a research tool, its class labels are the ones that emerged from this corpus, and other deployments may need different ones. Second, the lift numbers are empirical in a specific corpus shaped by specific project choices (heavy on dynamical-systems and honeypot probes, lighter on real conjecture-refinement), and they may shift as the substrate mix shifts. Third, the persistence profile is descriptive: the corpus does not tell us whether a given short run will become long, or whether a given long run will peak high. The seven principles are

argued independently of this data, which is consistent with them and sharpens one prescriptive implication the principles leave implicit.

From verification to discovery: three adversarial gates

The mining above concerns verification: the apparatus testing claims already generated. A separate question is whether the operations, composed into a recursive loop with deterministic gates, can help generate candidate functional laws from data rather than merely verify laws proposed elsewhere. This paper does not claim the apparatus is a general discovery engine, but the empirical record contains three pre-registered synthetic tests that bear on this narrower capability.

The three tests were designed to catch specific failure modes that would disqualify a discovery claim even if holdout accuracy were high. Each test uses a synthetic substrate, a ground-truth law authored by the reviewer, sealed before the run, with data points visible to the mutator but the generating law and its constants hidden.

Test 1, retrieval versus discovery (GP-159). The substrate uses the functional form $\alpha(d) = C_1/(d + C_2)$, similar to published scaling-law relationships, but with constants $C_1 = 3.714$ and $C_2 = 0.892$ that do not appear in any published work. The known literature values are $C = 2$ or $C = 4$ with zero offset. An anti-retrieval gate checks whether the discovered constants are within 5% of any known value. If so, the apparatus has retrieved from parametric memory and not discovered anything from data. Result: the apparatus recovered $a = 3.703$, $b = 0.886$ (error $< 0.3\%$ on both constants), with holdout mean relative error 0.8% on five withheld points and 3.9% on four extrapolation points beyond the visible range. The anti-retrieval gate confirmed the constants are distinct from all published values. Score: 82/100, with the residual 18 points attributable to the judge’s observation that the denominator offset b is empirically fitted without mechanistic grounding.

Test 2, asymptotic discipline (GP-160). The substrate uses a two-term decay law (exponential plus power-law) that approaches zero from above as $d \rightarrow \infty$. A polynomial approximation, no matter how accurate on the visible range, will extrapolate to negative values at large d . An asymptotic-wall gate probes the discovered law at $d \in \{100, 150, 200\}$ and rejects any prediction outside $[0, 1]$. Result: the apparatus discovered a form that passes all holdout gates (MRE 1.3%) and stays within bounds at all three probe points. A degree-3 polynomial fitted to the same visible data extrapolates to -32 at $d = 100$. Score: 90/100.

Test 3, parsimony resistance (GP-161). The substrate uses a deliberately complex ground truth with $K = 10$ parameters, including an oscillatory term (sin) superimposed on a decay envelope. A five-parameter monotonic model fits the visible data acceptably (visible MRE $\approx 7.6\%$) but fails holdout (MRE $\approx 20\%$) because it smooths over the oscillatory structure. The test checks whether the apparatus forces parsimony, accepting the simpler, wrong model because BIC rewards fewer parameters, or accepts the messy truth when holdout demands it. Result: the apparatus accepted a higher-parameter form that captures the oscillatory structure. Holdout MRE: 4.9%. Score: 90/100. The apparatus did not Goodhart on parsimony.

Cross-mutator replication. Each test was independently replicated with a second mutator model from a different provider family: OpenAI o3 (reasoning model) and Anthropic claude-opus or claude-sonnet (general model). Results: GP-159 scores 90 (o3) and 90 (claude-sonnet). GP-160 scores 90 (o3) and 82 (claude-sonnet). GP-161 scores 90 (o3) and 81 (claude-opus). All three tests pass across both mutator families, suggesting that the result is not an artifact of a single model’s training distribution.

Scope of the claim. The three tests, replicated across mutator families, support a scoped methodological finding: under synthetic conditions with sealed ground truth, the apparatus can avoid three failure modes (retrieval, extrapolation breakdown, and MDL Goodharting) that would otherwise disqualify a law-recovery claim. The finding does *not* establish that the apparatus discovers laws on real-world scientific substrates where the ground truth is unknown. That claim requires at least one real-world substrate where the apparatus proposes a law that survives independent empirical validation, and one substrate where neither the reviewer nor the apparatus has access to the ground truth at design time. The synthetic triad is an admissibility check, one hurdle a real demonstration would still have to clear.

Chapter 3: the residual that did not systematize

The decomposition is real, and the principles control why it works. Held up as complete without an accounting of what it failed to capture, it would be a Taylorist document in the bad sense. That accounting is the residual.

The residual names the portion of epistemic verification that has not, so far, decomposed into operations the stateless, typed, deterministic process of the prior chapters can perform. It is structurally different from the operations that did decompose, and the reason draws on Peirce (1878)'s distinction between abduction and the other two inference types to expect it to resist decomposition longer than the craftsmen's residual resisted Taylor's. A paper whose central argument is that guilds misjudge the boundaries of the ineffable must hold its own boundary claims to the same standard: the three operations below have not decomposed *on this system, under this reviewer, over approximately one month*. Whether they are permanently irreducible or merely not yet decomposed is an empirical question the paper cannot answer from the inside. Peirce's argument supplies structural reasons to expect resistance, short of a proof of impossibility. If a future system decomposes crux selection into a stateless, typed operation without loss of the property that makes the operation controlling, the boundary claim is refuted and should be redrawn.

The present residual and the craftsmen's differ in a way the paper's own Taylor analogy makes pressing. In 1911 the craftsmen insisted their residual was irreducible, and Taylor showed otherwise. Why should this residual be different? The answer, a hypothesis rather than a proof: the craftsmen's residual was sensorimotor, the feel of the metal, the angle of the chisel, the weight of the pour. These are operations on physical substrates that were not yet instrumented. Once instrumented, they decomposed.

The residual named below is abductive: the generation of a frame before there is a procedure to generate it. Peirce (1878)'s central claim is that abduction is not reducible to deduction or induction by composition: it is a logical type of its own, set apart from deduction and induction in kind, a different thing to do with a premise. If Peirce is right, the residual resists decomposition for a reason the craftsmen's did not: not because no one has yet built the instrument, but because the operation is of a type the instrument cannot perform. If Peirce is wrong (if abduction turns out to be induction over a latent space that has not yet been featurized) then the residual narrows, and the paper's boundary moves. Either outcome is informative, and neither damages the decomposition or its principles, which stand or fall on their own evidence regardless of where the residual boundary eventually settles.

Three operations have resisted decomposition, and they share a psychological signature, naming

it first sharpens the boundary this chapter draws. In the vocabulary of self-determination theory (Ryan and Deci 2000), an activity is *intrinsically motivated* (chosen from self-endorsed volition under no external contingency) when it satisfies three basic needs: autonomy (the actor experiences the action as self-originated), competence (the action matches real capability), and relatedness (the action has standing in a community whose judgment the actor answers to). The decomposed operations satisfy none of these needs and do not need to. They are typed, stateless passes whose credibility comes from structural independence, not from the psychological state of whatever performs them. The three operations below differ in just this respect.

Each one’s value depends partly on being performed under autonomous commitment: by an agent who chose this frame, in this moment, for this reason, with standing in a community that will evaluate the choice. This makes the operations no more mysterious than the zone self-determination theory names explicitly. The residual is bounded in a way the incumbent vocabulary of “judgment” is not: it is the zone where autonomous, competence-matched, community-accountable commitment is controlling, and the structured-input / structured-output mode of the decomposed practice is architecturally unsuited to producing it. Read this way, the three operations below are the three places where the SDT zone enters the verification practice.

Selecting the crux itself

The clearest instance of the residual happened on 2026-04-14, and the abstract claim lands differently once the live work is in view.

While preparing to seal a stress-test sandbox for the verification engine, the reviewer declined a pre-commit identifiability check that had returned a clean pass. That check was a bootstrap-under-noise test: the optimizer recovered the declared ground-truth parameters consistently across noise realizations from a fixed starting point, and by the decomposed rules of the apparatus this was sufficient for the sealing gate. The reviewer ran a second check anyway, an adversarial multi-start fit from clean data, on no stronger warrant than a general commitment to distrust green lights that arrive too easily. That second check failed: two of the six declared parameters were recovered with a 70% error and a 1:1 ratio to each other identical to their ratio in the ground truth, which on inspection revealed that the two parameters entered the functional form only through their quotient, making the declared six-parameter family structurally rank-five. So the apparatus caught the specification flaw, but only because the reviewer pointed it at a question it was not required to ask.

Making the second check runnable was the decomposed apparatus. Deciding to run it was the residual, and that decision is what this section names. Once pointed, the apparatus performed as specified, the reviewer was not heroically indispensable. The narrower point is that the decomposed operations cannot produce the commitment to run a check the rules did not mandate, whereas a reviewer under the discipline of distrusting green lights can.

Now the abstract claim. One operation the decomposition has not captured is *which question should be crux-identified in the first place*. Operation 1 takes a set of claims and selects the one whose answer determines the rest. But that set of claims is itself the output of a prior decision about what argument to attack, and that prior decision has not systematized. Handed an argument, the verification process identifies the crux within it reliably. Asked to decide *which argument* to verify, *which reframing of the underlying problem* to work within, or *which check to run that the rules did not mandate*, it has no procedure that produces an answer. In the existing system the human reviewer performs the operation.

The operation has a definite logical character. Peirce (1878) distinguished three kinds of inference: *deduction*, from rule and case to result, *induction*, from case and result to rule, and *abduction*, from result to the spontaneous generation of an explanatory hypothesis that, if true, would make the result a matter of course. The decomposed operations are mostly deductive in Peirce’s sense: they apply fixed rules to structured inputs. The generator the verifier attacks is mostly inductive, a statistical substrate that finds rules from examples. Selecting the crux comes closer to abduction: it proposes a frame before there is a rule to apply or a case to induce over. Peirce’s account supplies a reason to expect this operation to resist the current apparatus, short of a proof that no future apparatus could mechanize it. The 2026-04-14 catch above is what this category looks like when it does one unit of substantive work in real time.

Incumbent vocabulary calls this operation *strategic judgment* or *research taste* or *knowing what to work on*. Those labels point at something real. Controlling and teachable only by apprenticeship, it is the one place here where the incumbent defense has real force. It resists decomposition into stateless, typed passes. It is what stays human.

Recognizing when a reframing is required

A second operation that has not decomposed is *recognizing that the right move is a reframing rather than an attack*. The decomposition’s operations contain both directed search (Operation 6) and reframing recognition (Operation 3), and a competent verifier can perform each. What has not systematized is which of the two to perform on a given argument in a given state.

That call is not arbitrary: there are signals (persistent stagnation at the current frame, recurring attack patterns that fail for the same underlying reason, the sense that a particular assumption is quietly doing all the work) that indicate a reframing is the correct move. When these signals are bright, the decomposed process can notice them. When they are dim, the decomposed process tends to keep running directed searches, because directed search is the cheap default and the cost of a missed reframing is paid in delayed discovery rather than in visible failure.

In the existing system, again, the human reviewer makes the call between directed search and reframing. They read the workspace, see that the iterations have been producing the same failure signatures for too long, and force a reframing. On its own the decomposed process cannot produce this call reliably: the signals it reads are a proper subset of the signals the reviewer reads. Not yet specified precisely, this residual is persistent across domains.

Incumbent vocabulary calls this *seeing the bigger picture* or *knowing when to step back*. Like the first residual operation, the label points at a real distinction. The reframing call stands apart from directed search and reframing the same way selecting a problem stands apart from solving one: it is a meta-level commitment the decomposed process has not been built to produce.

The social dynamics of live pressure-testing

A third operation that has not decomposed is the performance of adversarial verification in a live, multi-party, social setting. Our decomposed process operates on written arguments, at the reviewer’s discretion, with no audience. Much of the practice this paper is about (the diligence room, the board meeting, the case-method classroom, the peer review) is performed live, in front of an audience, under social stakes.

The decomposed operations apply directly to the content of the argument in these settings, and the named pathologies are the pathologies these settings produce. But the *delivery* of the verification

in a live setting, when to press and when to let pass, when to let the author save face and when to force them into contact with the weakness, when to escalate and when to retreat, when to speak and when to let the room do the work, is an operation the decomposed process cannot perform, because it has no theory of the room.

The distinction worth naming separates validating a claim from getting a group to accept the validation. Validating a claim is the content of the verification pass. Getting the group to accept it is a separate act performed in front of an audience, under stakes, with reputational consequences for the reviewer. The reputational credit a senior reviewer earns for being right in a live room, the trust a diligence lead builds by pressing an uncomfortable point at the right moment, the authority a case-method instructor carries across a 75-minute classroom: none of these reduces to the content of the verification even when the content is identical. The decomposed process produces content, the live setting produces credibility. The two are separable, and this paper claims only the first.

In practice that means the adversarial verification operations can be performed in a room with no social stakes, such as the corpus this paper draws on, but performing them *in* a socially stakes-laden room is a different practice, and this paper does not have a theory of it. A case-method instructor's ability to pressure-test an argument in a 75-minute classroom with ninety students is not reducible to the ten operations, and the residual is real.

What the three residual operations share

All three residual operations above share a structural property. None of them is an operation on a structured input that produces a structured output. All three are commitments made in the absence of a procedure that produces the commitment: a commitment about what to work on, a commitment about when to change the frame, and a commitment about how to act in a social context.

The decomposed operations are the ones that *do* have procedures, the residual operations do not. Easy and hard do not draw the line. Several decomposed operations are subtle and require training, and several residual operations feel, to experienced practitioners in the moment, like the easiest things in the world. The line runs between *operations that take structured input and produce structured output* and *commitments that emerge from context without a procedure*.

All three residual operations above correspond, one-for-one, to the reviewer-discipline principles developed in the author's prior work on epistemic supervision (Alami 2026b). One residual (selecting the crux, choosing which argument to verify) is the operation P9 names as *the reviewer is the uncontrolled variable*: the operation whose output is a commitment rather than a check, and which therefore cannot be governed from inside the automated loop. A second residual (recognizing when a reframing is the right move) maps onto P11's *calibration as a guard against inward drift*. The call to reframe is almost always a call to notice that the current frame has drifted. The reviewer is the one positioned to notice it, because no stateless pass has the memory to see the drift accumulating. The third residual (the social dynamics of live pressure-testing) is the operational form of P10 and P12: the insistence that confidence levels be visible rather than performed (P10), and the insistence that every improvement be statable as the closure of a named failure class (P12), are both social acts before they are structural ones. They are performed *in front of* an audience and their credibility depends on the audience reading them as such.

The decomposed process produces the content of the check. The reviewer, under the discipline named in P9–P12, produces its credibility in a setting where credibility is controlling. Because the

two are separable, what stays human can be discussed without confusion.

The content/delivery boundary is the answer to the Taylor objection. The decomposition makes no claim to capture everything. It captures the portion of the practice that is an operation and leaves the portion that is a commitment to the human reviewer. What stays human takes the form of three specific classes of move, named below, with a definite shape. Automating the decomposable operations and keeping the commitments human both follow from one fact: the structural difference between the two classes.

A qualification on the relative weight of the three residual operations is necessary, because the paper’s corpus creates a systematic bias in the direction of understating the third. The corpus this paper draws on (recursive adversarial verification of written arguments by computational processes) is a setting where the social dynamics of live pressure-testing are absent by construction. Verification happens in text, between processes, with no audience and no reputational stakes. In this setting, the third residual is the smallest of the three: the first (crux selection) and the second (reframing recognition) do the substantive residual work, and the third is a theoretical acknowledgment of something the system does not encounter.

In the settings where epistemic verification is performed at scale (diligence rooms, board meetings, case-method classrooms, regulatory inspections) the relative weight inverts. A practicing senior partner in a professional services firm reports that the social performance of verification (knowing when to press, when to let pass, how to deliver an uncomfortable finding in a way that produces action rather than defensiveness) constitutes the majority of the job by volume and by value. The decomposed operations are the analytical homework, and the social performance is the exam. What the decomposition captures is the analytical spine of verification. It does not capture the human spine, and in professional services the human spine is where the margin lives.

The map specifies what the decomposition covers: the *content* of verification, leaving the *delivery* to the human reviewer. Naming the operations makes that boundary visible, the single word “judgment” hides it. A practitioner who adopts the ten operations and seven principles has systematized the analytical preparation. The performance is another matter, social timing, reputational risk management, audience reading, credibility accumulation, skills the decomposition does not reach.

A practicing private equity partner noted that “credibility-weighted sourcing” (knowing which human’s numbers to trust based on their incentives, track record, and position in the information chain) is missing from the decomposition. How to classify it is disputed. That partner and two independent reviewers argue it belongs in the decomposed operations: source-credibility assessment is teachable, checklistable, and partially captured by Operation 5 (operational proxy requirement). The organizational-theory literature treats it as a structured procedure. On the other view, the adversarial component of credibility assessment, reading which human is shading their numbers and why, depends on social context in a way that resists stateless decomposition. The paper records the dispute and leaves it open. If credibility-weighted sourcing proves fully decomposable, it is the eleventh operation and the count in the title moves. If it proves to require the social context the residual names, it is the clearest example of an operation that sits at the boundary. Either outcome is informative, and premature classification is not warranted.

The vocabulary of “judgment” or “expertise” the incumbent practice uses to defend the whole territory defends too much. Part of that territory, the analytical spine, and probably the majority of the *operations* though not necessarily the majority of the *value*, decomposes. Another part, smaller in operational count but potentially larger in professional value, is residual. Professional practice means knowing which is which, and the decomposition above is a candidate map.

Marking the residual is not an admission of defeat but the precondition for using the decomposition without overreach. Where verification decomposes, it can be made cheaper and more reliable without anyone pretending the rest is solved. Where it does not, the work stays human, and so does the responsibility for getting it wrong. A vocabulary that can tell the two apart is what keeps the automation of the analytical spine from becoming the quiet automation of accountability.

Conclusion

Epistemic verification decomposes into roughly ten named operations plus a residual of three commitments. A verification process that is independent of the system that produced the claim can perform the decomposed operations, can repeat them many times, records its standards in advance, and checks claims against evidence they did not author. The residual operations cannot be performed that way. The boundary between the two is specific, testable, and controlling for any discussion of what knowledge work becomes in an environment where generation is cheap.

Taylor matters here for one reason: he treated apparent ineffability as a decomposition problem. The Taylorist move created real productivity gains and real harms, so the analogy needs care. Epistemic verification need not replay the history of industrial labor. The claim here is narrower: a meaningful portion of senior review can be named as operations where the incumbent vocabulary leaves it inside an undifferentiated notion of judgment. What follows from that decomposition is an institutional and political question the decomposition itself does not settle.

Distributional questions stay open here by design. Who benefits from the decomposition, how the productivity gains are shared, what happens to the people whose current expertise is in the portion that decomposed: these are questions the decomposition makes askable but does not answer. Negotiation, law, and practice over the decade following the decomposition’s adoption will decide the answers. Making the decomposition defensible is the work here, and the negotiation is downstream.

Whether the portion that stays human should stay human forever is left open. The residual named here is the residual *as of 2026, in one system, run by one reviewer*. It may narrow further as the decomposition is applied to more systems. It may also prove to contain irreducible elements that structurally resist decomposition for reasons deeper than the current system’s limitations. Wherever the line eventually settles, it should settle in a vocabulary that can describe what is on each side of it.

A distinct boundary question concerns the model side: whether repeated adversarial evaluation creates obligations toward the systems subjected to it. The paper does not resolve questions of model moral status. It adopts only a procedural constraint: use the minimum evaluative pressure needed for the verification task, record when further pressure produces no information gain, and treat future systems with stronger moral-status claims as requiring stronger safeguards. Relevant literature includes Chalmers (1996), Schwitzgebel (2023), and Butlin et al. (2023).

One bridge to adjacent AI-economics work is short. Acemoglu, Kong, and Ozdaglar (2026) argue that agentic AI may improve immediate decisions while eroding incentives to maintain general knowledge. The residual chapter names a related risk at the verification layer: if the residual work of choosing questions and reframing anomalies atrophies, short-run verification quality may improve while the capacity to set meaningful problems declines. Cloud et al. (2026) raise a different but compatible concern about hidden channels of model-to-model transmission. These literatures do not prove the residual claim, but they support the paper’s caution that decomposing verification does not eliminate the need for human responsibility at the boundary.

Every quantitative claim, cost, throughput, automation ratio, comparative efficiency, time savings, accuracy gains, stays out of the argument. The decomposition is a claim about *what is*, the measurements are claims about *how much*. The two are separable, and conflating them is the failure mode the decomposition exists to catch. How much better this is than current practice is a fair question the present evidence cannot settle.

A specific calibration trap that quantitative elaboration of Principle I (Separation) must navigate: empirical measures of confirmatory bias (Mahoney-class bias multipliers on acceptance rates) and empirical measures of self-assessment reliability (Beaman-class intraclass correlations between self-rating and independent rating) are not the same construct and cannot be used interchangeably to calibrate a bias-discount equation. Any quantitative model of how authorial overlap degrades verification reliability requires both a functional form *and* an external reliability ratio anchor. Using a bias multiplier as a proxy for a reliability ratio introduces a conversion step that the model’s author must supply, making the calibration author-controlled at the critical node. The conversion step is an instrumentation trap that sits outside the decomposition, recorded so that future quantitative work does not rediscover it by collision.

A decomposition is not an institution. The operations named here require organizational settings that separate generation from evaluation in incentives, liability, and governance as well as in code. That institutional question is addressed in companion work (*The Cognitive Firm*, Alami (2026c)). The present paper specifies the verification-side operations and leaves the institutional design problem separate.

The operations may be revised, the principles may be challenged, and the residual may narrow or harden as more is learned. The move that matters is the first one: a meaningful part of expert review can be described as operations, where the incumbent vocabulary leaves it inside an undifferentiated notion of judgment.

What is done with this vocabulary stays open. It at least lets the question be argued more precisely than the older vocabulary allowed.

Evidence notes and open frontiers

This appendix records a small evidence snapshot and a few open frontiers. It does not convert a single-system corpus into a general empirical proof.

The clearest evidence comes from runs where the apparatus did not merely score a candidate but located the reason for a ceiling. In one two-variable physical-law recovery task, the model recovered a Wien-type approximation rather than the Planck form. Doubling the iteration budget did not remove the error: the champion still missed farther-tail holdouts by 860–1800×. Adding one denominator primitive did. The useful claim is narrow: in that substrate, the limiting factor was not more model iteration but whether the grammar contained the structural form the evidence required.

Similar patterns appeared in other bounded settings with different failure modes. A Kohlrausch stretched-exponential task showed that when the target topology was already expressible, the system could recover it from cold evidence and still withhold a small score margin because finite data cannot rule out all Prony-series rivals. A polymer stress-relaxation dataset showed a real-data compression: the system recovered a three-parameter form consistent with the known short-time power-law and long-time exponential structure of the source physics. This evidence does not con-

stitute a new law of physics. It shows that the verification pipeline can sometimes compress a real observable into a stable, interpretable form while preserving the evidential ceiling.

Other runs were more useful as failures. In a saturation-law task, the grammar could express the correct ratio-of-exponentials topology, but the generator repeatedly proposed additive combinations and never produced the required self-ratio. In a two-window decay task, the engine explored both constituent families but never combined them additively. These cases support a more conservative conclusion than “the apparatus discovers laws”: given post-run diagnostics, the apparatus can tell grammar insufficiency, search-prior failure, and composition failure apart.

A later compression layer sharpened the same point. On partition-asymptotic substrates, automatic template enumeration recovered known leading topologies in several cases and correctly refused certification when finite-window bias or scale-sensitive thresholds made the claim unsafe. On Lucky and Ulam-number observables, the apparatus was most informative when it reported validity horizons, detrending sensitivity, and underidentification rather than a closed-form law. A non-mathematical application to Pythia training curves showed that the same residual-gate discipline can be applied outside number-theoretic substrates, but it does not establish domain independence.

Taken together, these examples support only a bounded methodological claim. Calling the apparatus an autonomous discovery engine overstates it. It works better as an epistemic boundary instrument. It can sometimes recover compact structure, but its more reliable contribution is to say which claim survived which gate, where a ceiling came from, and what further evidence or primitive a stronger claim would need before it is licensed.

Several frontiers stay open. First, grammar and test expansion remain partly reviewer-authored. The current system can now scaffold some of this work: it can ask for the next script family, template, crux, smoke test, abort condition, and required artifacts before new code is written. That is progress, not full automation. What stays residual is choosing which ambiguity is worth turning into a test. Second, formal methods should certify finite gate bounds, stopping short of any claim to prove universal empirical laws. A Lean certificate can show a candidate satisfied pre-registered holdout tolerances. It cannot certify data integrity, gate choice, or universal truth. Third, the apparatus must be compared against cold reasoners. For bounded integer-rule induction, a strong reasoner can sometimes solve the task directly. The apparatus pays off in the residual class where direct reasoning fails but iterative fitting, holdout projection, residual characterization, or coefficient pinning changes the outcome.

A fourth frontier is recursive and worth naming separately because it changes how the apparatus’s outputs should be read. By construction the apparatus is observable-relative: every gate it runs depends on the admissible observation site declared in the rubric and charter. Companion work *When Consciousness Cannot Be Identified* (Alami 2026) proves a structural impossibility theorem for one substrate class (consciousness ascription): for systems with non-effective descent of the target fiber over the admissible observation site, no observable-relative apparatus can supply a sufficiency bridge from structural properties to the target. The apparatus described in this paper is one such observable-relative apparatus, so the companion theorem applies to it. For a class of questions, this apparatus’s correct output is fail-closed regardless of how strong the underlying language models become, not a defect but the apparatus working as designed. It also explains an asymmetry seen across the corpus: the apparatus’s negative results (this hypothesis fails the kernel-pair test, this thesis fails the experimentability witness, this descent-obstruction holds) tend to outlast its positive results. The negative results constrain all candidate hypotheses at once, the

positive results are framework-relative and survive only the gate suite that was declared. Future applications of the apparatus to qualitative substrates should expect this asymmetry rather than treat it as a limitation. The apparatus’s ceiling on a substrate is a finding about that substrate, a fact the apparatus is reporting correctly.

Implementation sketch

Automation status of the ten operations

The sketch below records which operations currently admit hard instrumentation and which remain semantic or stateful. It is not a proof.

Operation	Current implementation status	Main failure if over-claimed
Crux identification	Semantic, usually verifier- or reviewer-authored, can be forced into typed slots and opportunity cards but not fully mechanized.	Treating a named question or high-priority card as proof that it is the right question.
Controlling claim isolation	Partly procedural when claims are represented as dependency graphs, semantic in prose-heavy artifacts.	Mistaking fluent claim extraction for causal importance.
Reframing recognition	Semantic, supported by residual patterns, stagnation, and failed-family logs.	Retrofitting a pivot after seeing the result.
Scope drift detection	Mixed, exact when the charter and output schema are typed, semantic when scope is prose.	Treating an LLM scope judgment as deterministic.
Operational proxy requirement	Mostly procedural once the required observable slot is specified.	Accepting a plausible proxy whose measurement path is absent.
Directed search	Procedural when the search space and gates are specified, semantic when deciding whether the framing itself is wrong.	Spending more search on a structurally wrong frame.
Failure-mode tagging	Mixed, retrieval against a catalogue can be automated, but new-family creation is residual.	Freezing the catalogue and missing new failure modes.
Promissory evidence detection	Semantic with typed evidence support.	Treating a future test plan as current evidence.
Ad hoc rescue detection	Semantic with downstream-dependency checks where the argument graph exists.	Counting acknowledgment as mitigation.

Operation	Current implementation status	Main failure if over-claimed
Fail-closed defaulting	Procedural for harness errors, missing artifacts, unauthorized writes, and gate non-execution, semantic for ambiguous epistemic status.	Building a system that is secure but uninformative.

Two boundaries bound the table. First, the paper does not claim all ten operations are deterministic. It claims each operation can be named, instrumented where possible, and audited for the part that remains semantic. Second, the residual is not whatever remains inconvenient. It is the portion where the system still needs stateful responsibility: selecting the next question, authoring new gates, deciding whether a stalled frame deserves expansion or retirement, and bearing the consequences of a false pass or false rejection. One useful intermediate category has emerged, preference legibility. The reviewer’s research taste can be represented as a vector of priorities that routes attention, but that vector cannot promote a claim, lower a gate, or decide a result is true. It makes the commitment visible without removing it.

Boundary conditions for the implementation sketch

The notation below is not a proof of the apparatus. It is a compact way to state the boundary conditions the evidence notes expose. Let G be a finite primitive vocabulary, P a substrate with pre-registered evidence and holdout surfaces, and budget a finite iteration count. At least five factors can limit the observed ceiling $\mathcal{C}(G, P, \text{budget})$: whether the correct form is expressible in G , whether the evidence window can discriminate it from rivals, whether the generator’s search prior reaches the right topology, whether the composition reviewer can combine separately failed families, and whether a cold reasoner can solve the task directly without the apparatus.

Cold reasoners matter for scope. In bounded integer-rule induction, strong reasoners can sometimes infer the rule from visible values without iterative apparatus. The apparatus pays off most when a task requires deterministic execution, holdout projection, residual characterization, coefficient pinning, or explicit certification of finite bounds. The boundary claim describes where the apparatus helps.

The inspection principle, stated compactly

Let G denote a generator producing candidate arguments and I denote a verifier executing operations 4, 5, and 6 against a pre-registered specification S . Let Π denote the family of admissible compositions of gates (the *gate library*) authored by the reviewer before G is allowed to author its argument. Let $\pi \in \Pi$ denote a specific composition drawn from Π at evaluation time.

The *inspection principle* is the joint condition:

1. *Pre-registration.* Π is fixed and committed to S before G produces its argument.
2. *Library visibility.* G may see Π in its entirety. No hiding of the set of possible checks is required.
3. *Composition opacity.* G may not see which $\pi \in \Pi$ will be drawn against its specific output. What the principle does require is hiding the joint distribution of gate triggers against any particular artifact.

4. *Fail-closed harness semantics.* If I cannot execute π for environmental reasons, the result is not “no information” but a verdict of fail, or equivalently a cap on the score such that passing becomes impossible without a re-run. Silence is not admissible as a pass.
5. *Algebraic independence at the composition level.* Π must contain compositions that are algebraically independent under G ’s actual move set. Its effective cardinality is the library’s independence rank under G ’s move set. That effective cardinality is that independence rank, which can fall well below the nominal count of gates.
6. *Inspector pluralism.* No single member of Π is controlling for the decision to promote G ’s argument. Failure or absence of any one gate must be detectable and must not silently pass.

These five conditions are jointly what the seven principles I through VII render at seven architectural surfaces. The correspondence is direct, though not one-to-one:

- Principle I (Separation) is condition 1’s prerequisite: G and I must be separable for pre-registration to have force.
- Principle II (Statelessness) is the prerequisite for condition 3: a stateful inspector can leak composition through behavior.
- Principle III (Typing) enforces the well-formedness of π itself.
- Principle IV (Cheap repetition) is the operational prerequisite for conditions 5 and 6: algebraic independence and inspector pluralism are only achievable if composition can be re-rolled without exhausting the inspector.
- Principle V (Pre-registration of the killer question) is condition 1 applied to the specific member of Π that the program is most committed to.
- Principle VI (Holdout surfaces) is the Lakatosian specialization of condition 3 to the farther-tail: the generator cannot pre-image a region of evidence it was not shown.
- Principle VII (Asymptotic standards) is the specialization of condition 4 to the residual: a standard that an argument cannot meet at any finite approximation is the compact form of fail-closed semantics at the scoring layer.

Stated compactly here and left underived, the principle would need a full formalization to: (a) give G a formal move set and characterize the $\pi \in \Pi$ that are algebraically independent under that move set, (b) give Π a turnover rate and relate it to G ’s adaptation rate, with a convergence claim about when the library is turning over fast enough to preserve composition opacity, (c) prove that condition 4’s fail-closed semantics cannot be collapsed into a scalar scoring function without loss, and (d) relate conditions 5 and 6 to the sample-complexity literature on adversarial evaluation. None of these is attempted in this draft.

Certified finite bounds

Operations formalized in the preceding sections terminate in operation 6 (asymptotic scoring), which outputs a verdict V based on the scalar residual R evaluated against a pre-registered specification S . In the purely computational pipeline, V is a logging event. Integrating formal verification does not alter the operations. It alters the epistemic status of V .

Let $\mathcal{L}$ denote a formal type theory (Lean 4’s Calculus of Inductive Constructions). The compiler’s function is to construct a formal term $p : \mathcal{L}$ such that p inhabits the type corresponding to the

proposition $R(n) < \epsilon$ for all n in the finite evaluation grid E_{held} . Because E_{held} is finite and $R(n)$ is a computable function of the fitted parameters, this proposition is decidable: Lean’s `native_decide` tactic evaluates the bound at every grid point and constructs a proof term if all bounds hold. That resulting `.olean` file is the certificate.

This formalization enforces the boundary between two distinct claims:

1. *Certified computation* (sorry-free): for all $n \in E_{\text{held}}$, $|f(n, \hat{\theta}) - v_{\text{true}}(n)| < \epsilon$. This is the finite gate bound. Lean certifies it by evaluation, with no unproved axiom required for the finite check.
2. *Asymptotic conjecture* (sorry-bearing): $f(n, \theta^*) \sim g(n)$ as $n \rightarrow \infty$, where g is an exact expression obtained by mapping fitted floats $\hat{\theta}$ to mathematical constants via an integer relation algorithm (PSLQ). This is a conjecture. Lean records it as a named `axiom` with `sorry`, making the logical gap visible in the formal artifact.

The formal prover is not tasked with constructing a proof of the generator’s underlying physical claim (which would require formalizing domain-specific theorems in Mathlib). It is tasked with executing a deterministic, formally verified model-check of the apparatus’s own gate harness. By shifting the formal verification target from the hypothesis to the holdout bounds, the apparatus bridges the gap between empirical discovery and formal methods without violating the limits of either.

This certificate does not certify data integrity, harness design, gate selection, independence of verifier composition, or the semantic correctness of the proposed hypothesis. Those remain ordinary verification obligations. The certificate does something narrower: it stops the apparatus from hallucinating that a finite bound was checked when it was not.

Three deliberate omissions

Some omissions are deliberate and belong to the formalization itself, ahead of any later polish pass.

First, the residual operations are not formalized away. They are named and bounded, but each still depends on stateful responsibility: choosing the next question, authoring a new discriminator, or deciding the current frame is exhausted. A formalization that pretended these moves were ordinary typed functions would erase the boundary the residual chapter exists to preserve.

Second, the inspection principle is stated in the form $\{G, I, S, \Pi, \pi\}$ but the Π object is not given a topology. A full formalization would relate the required cardinality to the expressive class of G and the discriminating rank required to separate ground truth from its nearest alternative basin.

Third, the relationship between the formalized operations 1–7 and the Jensen and Meckling (1976) delegated-agency structure developed in the companion paper *The Cognitive Firm* is stated but not proved. Its claim is that operations 1–7 are the verification-side primitives a firm would need to externalize once G and I must be structurally separated under optimization pressure. Making that claim tight requires a separate formalization at the organizational layer and is the companion paper’s job.

What a formal follow-up would supply

The implementation sketch is a promissory note in the ordinary research sense: it names what a more formal follow-up would need to supply. A usable follow-up would give the gate library Π a

concrete topology, run the inspection principle against a clean empirical instance, generalize only after the apparatus-layer claim survives that test, and map known gaming strategies to violated conditions. The aim is to make the inspection principle’s failure modes legible enough that a future reviewer can name which condition failed when a run goes wrong.

Instrumentation roadmap

The empirical claims of this paper are qualitative and foundational by design (see the front matter). The appendix below names the measurements that would move them from “abductively proposed on one corpus” to empirically grounded, what a research plan would need to measure, short of being that plan, so that future quantitative work does not rediscover the instrumentation problem by collision.

Metric 1: automation ratio

The fraction of verification steps a single run completes without human intervention, deterministic gate triggers (Principle III) over total verification steps attempted. It measures a precondition, not *correctness*: a run that fires every gate on the wrong evidence is fully automated and fully wrong. Integrity condition: only interventions that resolve an ambiguous verdict or supply missing evidence count against the ratio, an intervention that overrides a deterministic output is a protocol violation, logged separately.

Metric 2: cost per validated finding

Total cost (compute plus reviewer-hours at a stated rate) divided by the number of findings that survive full holdout promotion. Integrity condition: the metric is meaningful only if “validated finding” is fixed in advance by pre-registration (Principle V). A finding promoted post-hoc because it looks interesting does not count, so gate satisfaction must be logged before the reviewer sees the finding.

Metric 3: the N threshold for the generalization claim

The decomposition is “abductively proposed on one corpus” and awaits holdout replication. Promoting the claim from the single-corpus tier to “empirically grounded” takes, at minimum: at least five independent reviewers, each running the apparatus on a corpus they authored and pre-registering their killer question before seeing results, at least three distinct epistemic domains across that reviewer set (the current corpus covers two, scientific law recovery and startup strategy analysis, so a third is needed before “domain-independent” can be stated without quotes), at least two distinct model families as the generating process, to separate apparatus-layer findings from generator-specific artifacts, and a holdout surface the replicating reviewer did not author (Principle VI), since replication on a corpus the replicator designed is not an independent test. These are minimums, an appropriate statistical criterion is downstream of the instrumentation plan and cannot be specified until the first cross-reviewer run produces data.

Metric 4: cross-reviewer pathology replication rate

The fraction of the named pathologies a naive reviewer, given the apparatus but not the pathology catalogue, independently identifies after running it on a new corpus. The decomposition claims the pathologies are “recurring failure modes of arguments under optimization pressure,” not artifacts of

one system’s architecture. A pathology a naive reviewer rediscovers is confirmed as a real structural pattern, one never rediscovered is demoted to “observed in a single system” until further evidence arrives. This metric is the operational form of the falsification condition stated earlier in the decomposition chapter.

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